



KRONOS
MECHATRONICS

PRIDE 2 – Übung

3D Printed Electronics via Additive Manufacturing

15.06.2026 - Johannes Hörber

PRINTING THE FUTURE IN EVERY DIMENSION.



Agenda



1. Wer ist Kronos Mechatronics und was machen wir?
2. Was ist 3D gedruckte Elektronik (3DPE)?
3. Wo kann man 3DPE einsetzen und welche Vorteile hat diese Technologie?
4. Wie wird 3DPE gefertigt? – Beispiele anhand der Kronos Plattformtechnologie
5. Wo liegen zukünftige Herausforderungen?
6. Besichtigung der Kronos Anlagen am Lehrstuhl FAPS – Laufender Druckversuch



KRONOS
MECHATRONICS



KRONOS Mechatronics pioneers **real 3D free-form electronics**, enabling conductive and functional structures to be printed and embedded directly into complex geometries - beyond the limits of planar electronics and conventional 3D printing

FOUNDED IN

2024

IN GERMANY

USING EXPERTISE AND ASSETS
ACQUIRED FROM

Neotech AMT
Advanced Manufacturing Technologies for 3D Printed Electronics

Team

IN NUEREMBERG & HAMBURG

WITH BACKGROUND IN

MANUFACTURING
ENGINEERING
NATURAL AND COMPUTER SCIENCE
ECONOMICS

Branches

COMMUNICATION

DEFENCE

AEROSPACE

AUTOMOTIVE

MEDICAL

INDUSTRIAL ELECTRONICS

WEARABLES & SMART TEXTILES

Business Focus

MACHINERY, SOFTWARE & SERVICES

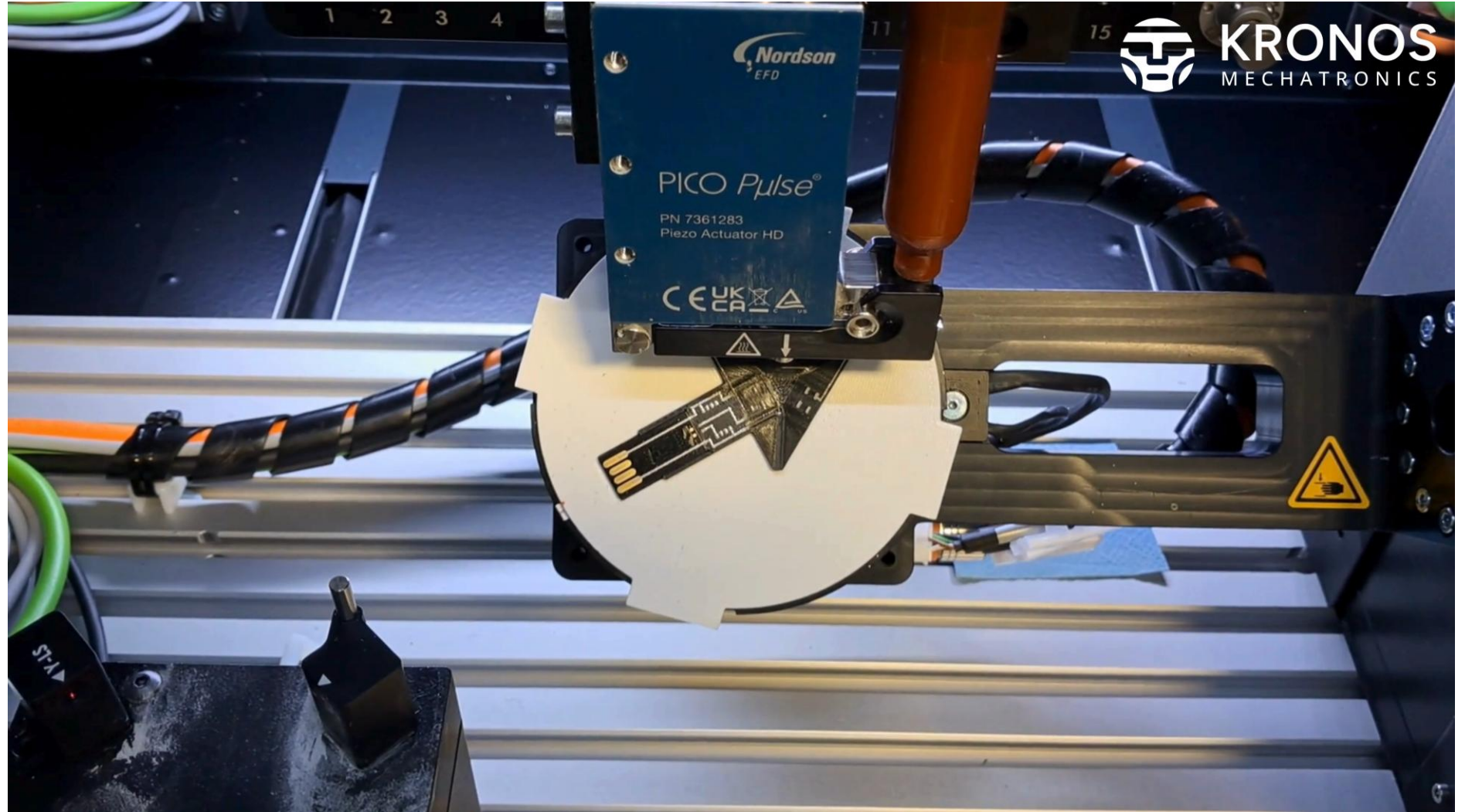




KRONOS
MECHATRONICS



at a glance:





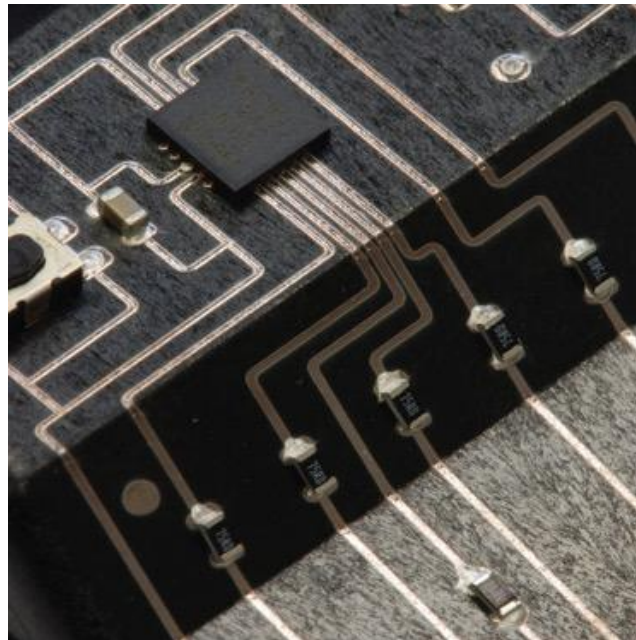
What is 3D Printed Electronics and where can it be used?

What is 3D Printed Electronics (3DPE) or Additively Manufactured Electronics (AME)?

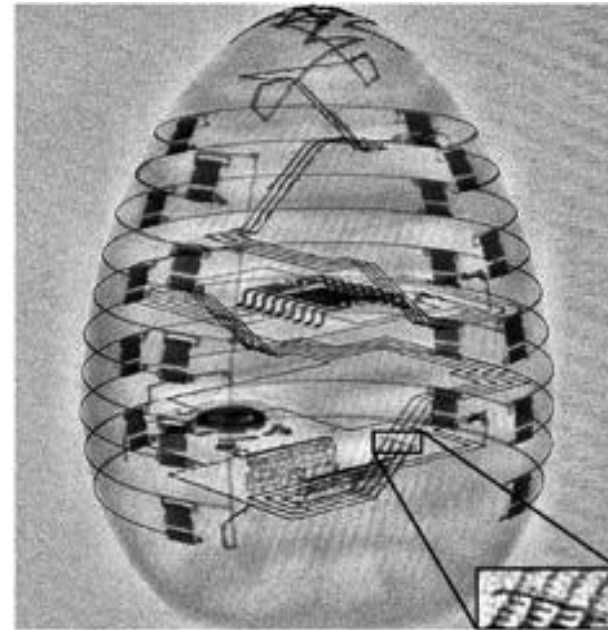


3D PE / AME is the addition of printed electronics functionality, sometimes in combination with classical SMDs onto and/or into structural components to create mechatronic systems

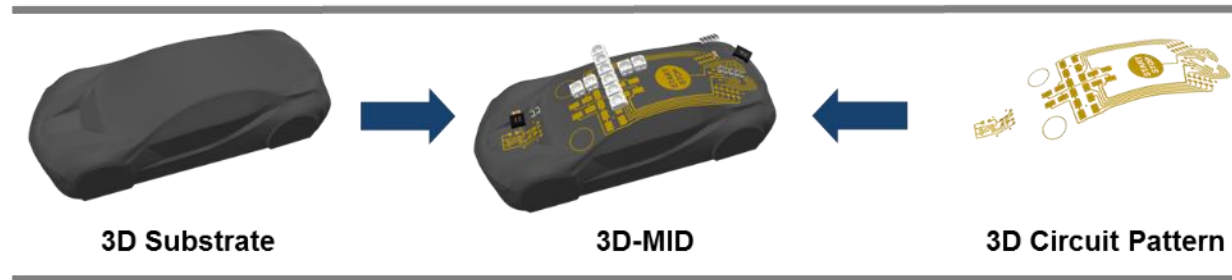
Printed Electronics on a 3D Substrate



Printed Electronics in a 3D Substrate

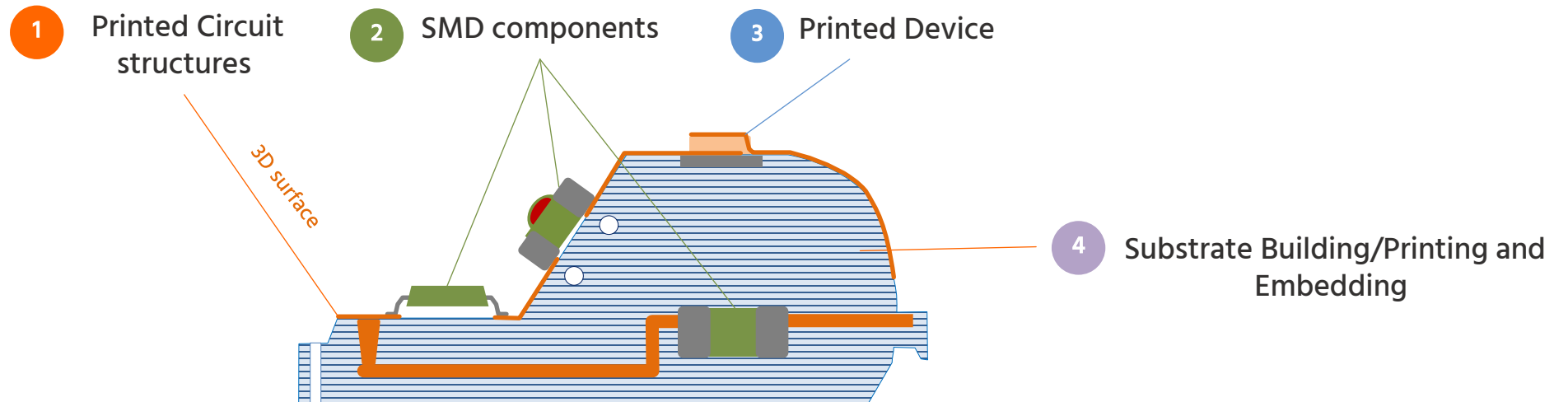


3D-MID enhanced by Printed Electronics and AM



Courtesy: Research Assoc. 3-D MID e.V.

combination with AM and
Printed Electronics

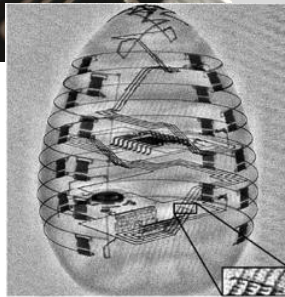
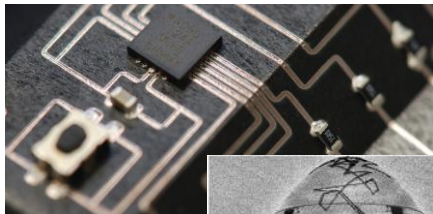


Functions Overview



Interconnection & Contacting

Conductor tracks on and inside components in combination with Pick & Place

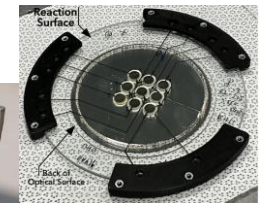
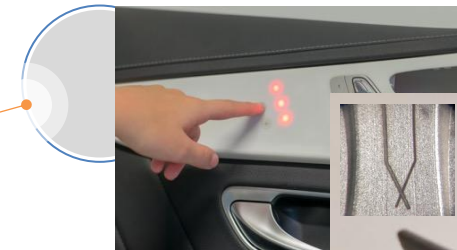


3D

Fully integrated digital manufacturing of mechatronic systems in a single machine unit

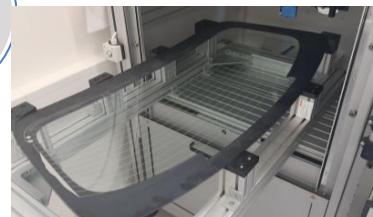
Actuator & Sensor Technology

Sensor arrays for touch sensors, temperature sensors, and electroactive polymers as actuators and sensors



RF - Structures

Antenna structures



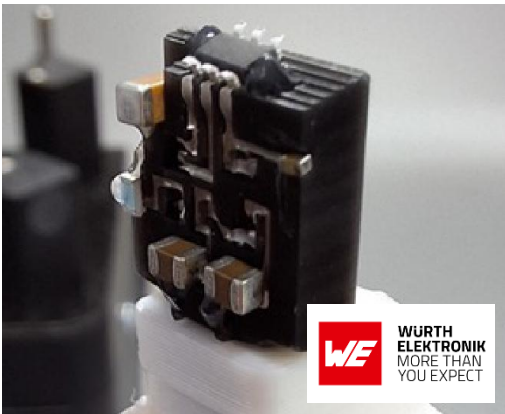
Heating structures

PTC heating structures



Combined heating/antenna structures

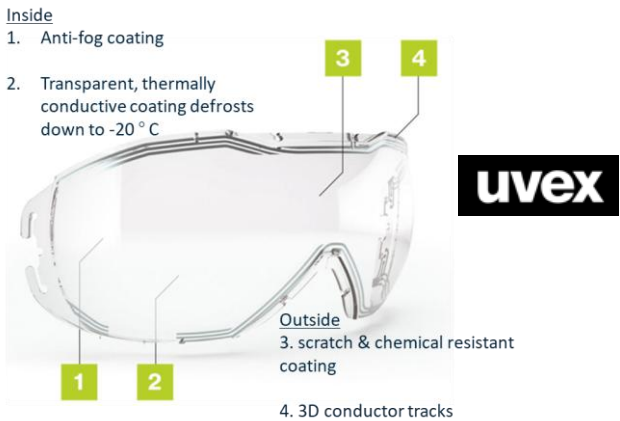
What are the Benefits – Application Examples



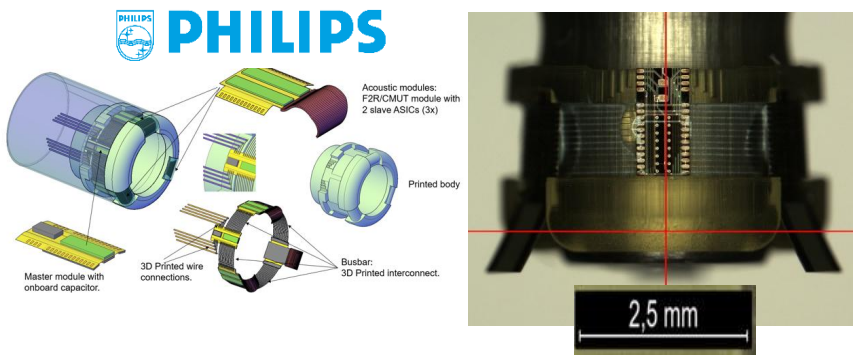
INDUSTRIAL ELECTRONICS



AUTOMOTIVE



WEARABLES & CONSUMER ELECTRONICS



MEDICAL APPLICATIONS



MOBILE COMMUNICATIONS



Application Examples – Printed Antennas

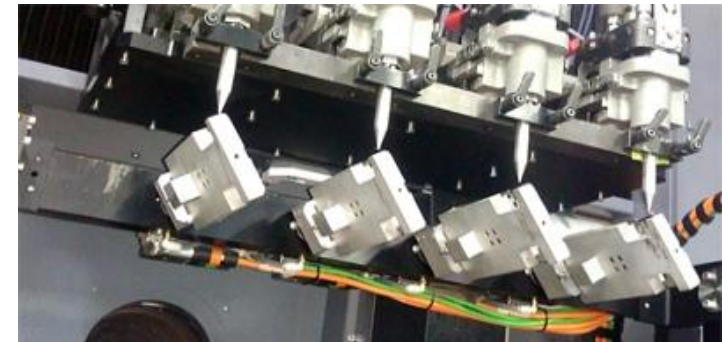


Current Process Route using Aerosol Jet Printing:

1. Printing of Ag-nanoparticle inks on filled PA resins and oven sinter
2. RF Performance: matches industry standard
3. Low temperature inks for PC/ABS
4. Production Costs: specific antenna designs show cost reduction of compared to current manufacturing techniques



Demonstration Antenna
Courtesy: LITE-ON Mobile Mechanical SBG



Multi-station Printing.
Courtesy: LITE-ON Mobile Mechanical SBG

Heated Goggles for PPE



Scratch-resistant and fog-/ice-free protective goggles with "built-in lens heating"

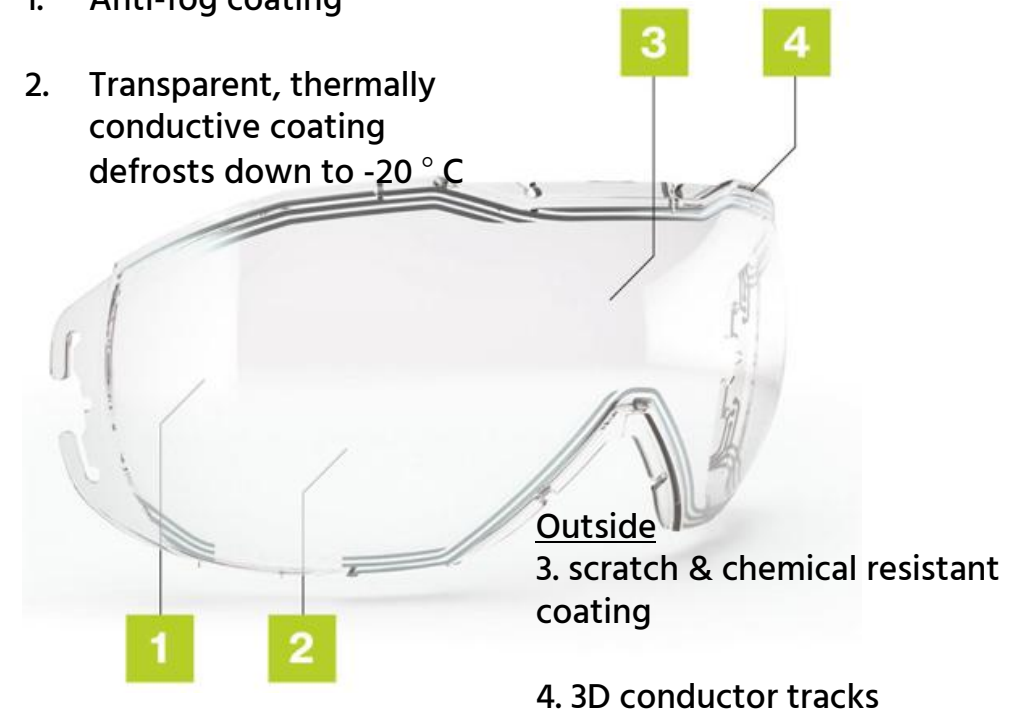
Unveiled at A+A 2021 - International Trade Fair for Safety, Security and Health at Work



Courtesy: UVEX ARBEITSSCHUTZ GMBH

Inside

1. Anti-fog coating
2. Transparent, thermally conductive coating defrosts down to -20°C



Single Pair Ethernet Connector



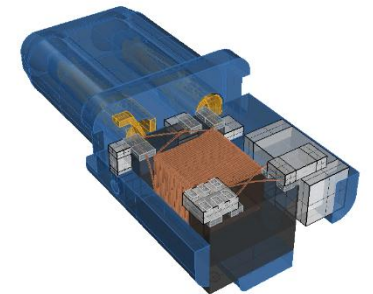
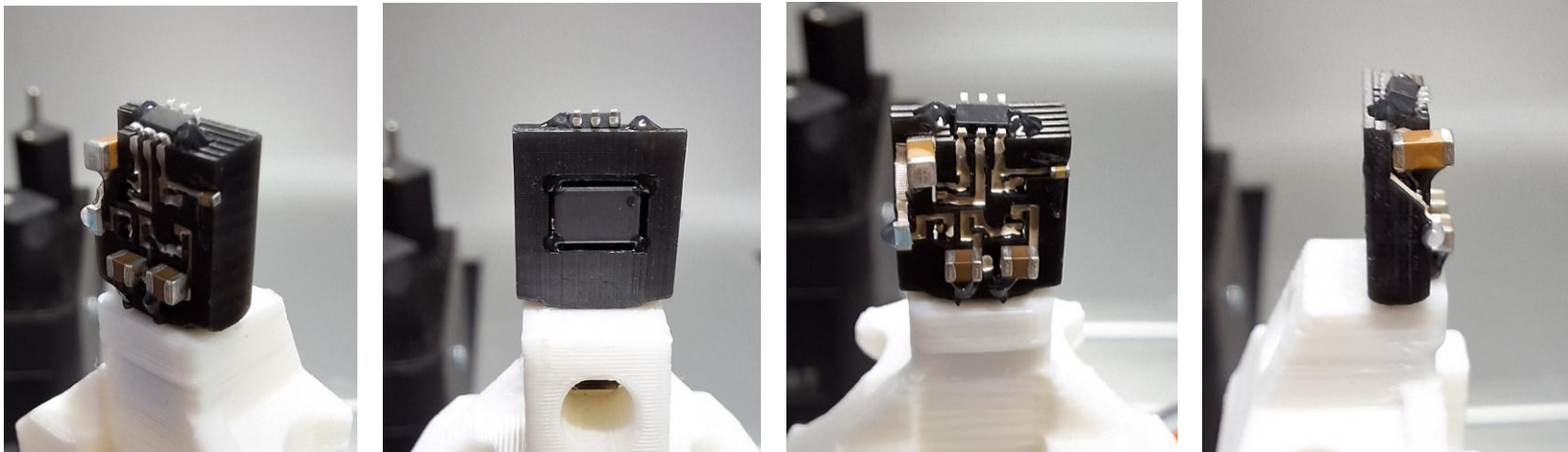
Leverage AM to enable agility: low volume-high mix.

Benefits:

- Miniaturisation
- Reduced parts count
- Agility & Customisation -> service premium segment
- Localised Manufacture



AMPERE

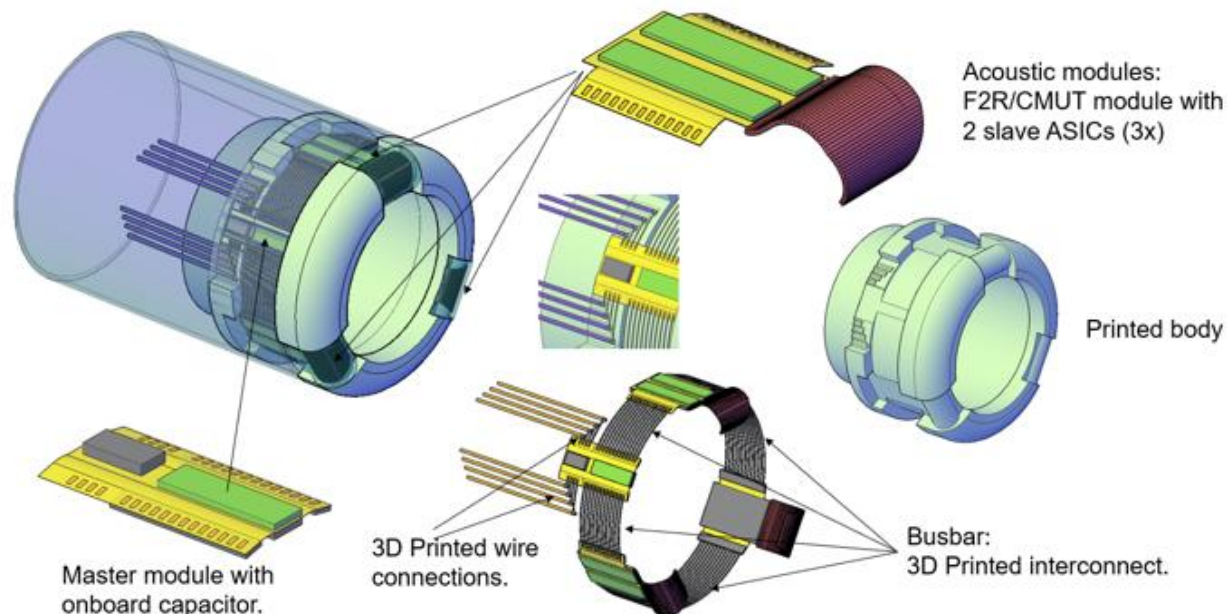


Lead Extraction Unit/“Ultrasonic Camera”



Fully Additive production enables:

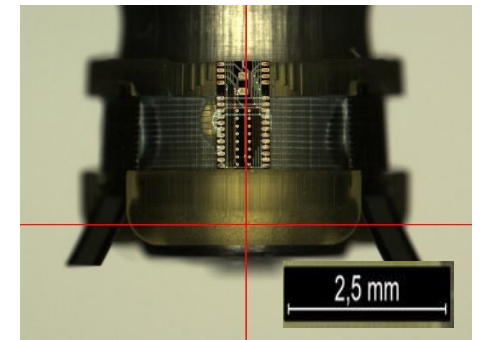
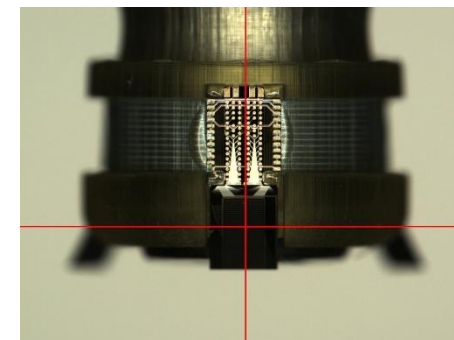
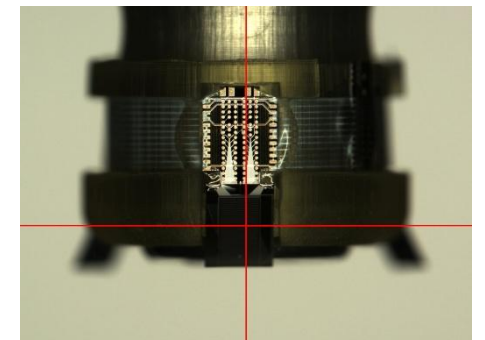
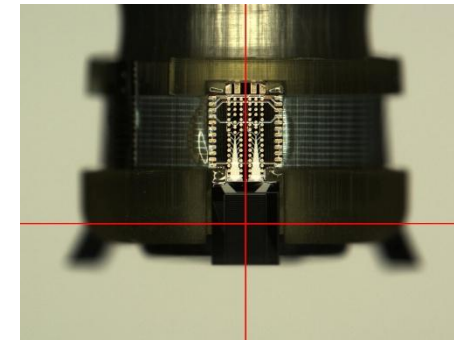
- Unique Design.
- Shorter design loops, reduced time to market.
- Easier to add functions & change design, add diversity.
- Less waste as design is changed.
- Less stock due to printing on demand Interconnected ($20\mu\text{m}$ line & Space) ASIC/CMUT on SLA printed body.



AMPERE



PHILIPS



Additively Manufactured Spot Light

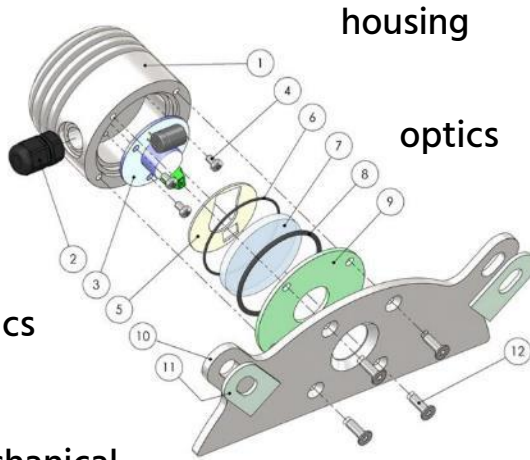


Example: Lighting Applications

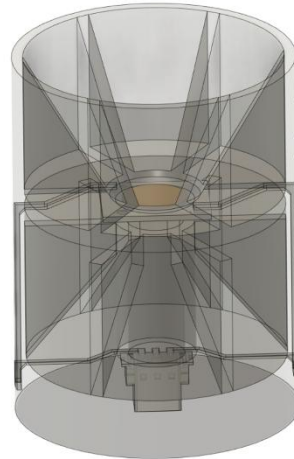
connection
to
peripheral
electronics

internal
electronics

mechanical
fixtures etc.

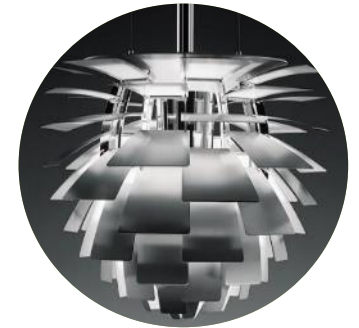


AMPERE



signify

combination/integration of functions
and parts consolidation



Typical product architecture

- large quantity of parts
- separate parts for separate functions
- complex design of interfaces (failure risk)

Part count reduction

- easy assembly
- simplified supply chain

Integration

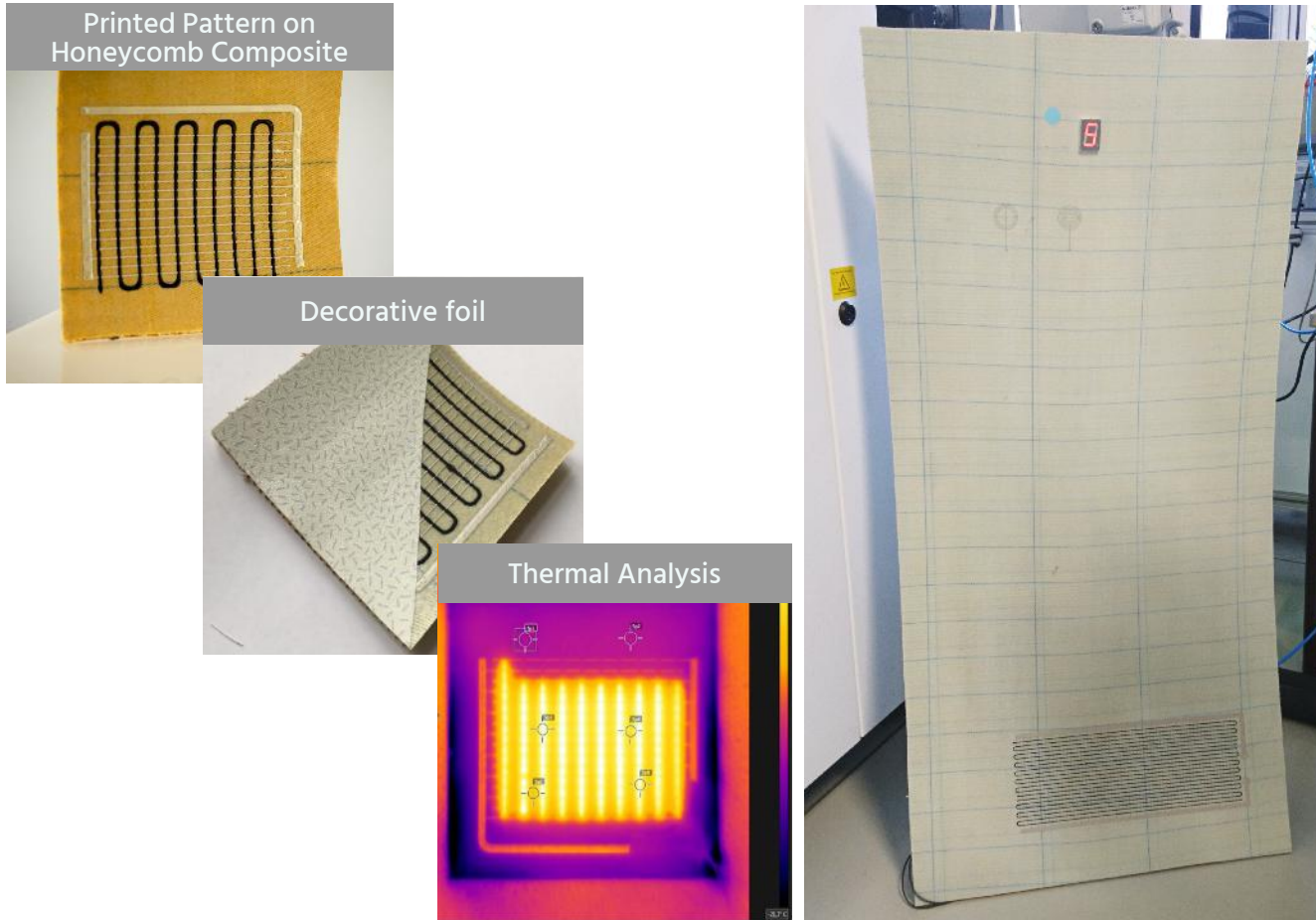
- miniaturization
- increased performance

Design and Customization

Large Area Printed Electronics



Multimaterial processing and Large Area Electronics:

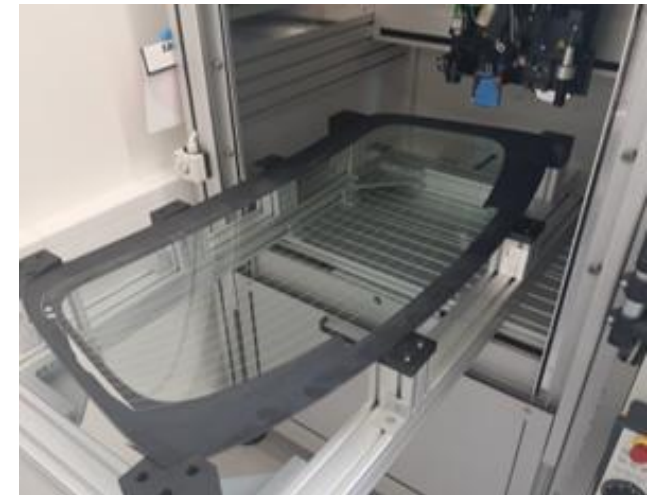


Benefits compared to flex PCB / foil:

- Complex shapes feasible
- Printing of vias
- Integration of mechanical and electronic components

Example Automotive application:

Wind screens with integrated heater structure and antenna pattern



Benefits - Summary



Design Flexibility

- Integrated Products
- Flexible Form Factor
- Miniaturisation
- New Functionality

Economics

- Reduced Parts Count
- Materials Efficiency
- Shorter Process Chains
- Automated Manufacture

Sustainability

- Reduced Materials Mix
- Additive Manufacture
- Simplified/Automated Recycling
- Localised Manufacture - Onshoring

Macro trends driving technology adoption:

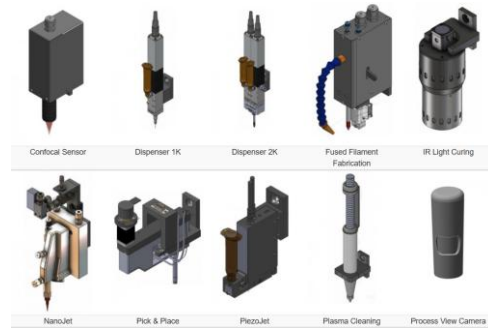
1. Sustainability – move from wasteful wet chemical subtractive manufacturing to more resource efficient, clean AM.
2. Electromobility - switch from fossil fuel transport. Opportunities from electrification and weight reduction requirements.
3. Industry 4.0 & IoT. Products with added intelligence and novel design.
4. Onshoring - Fragility of current electronics supply chains – Localised manufacture.

Fully expect 3D PE to mirror growth of classical AM.

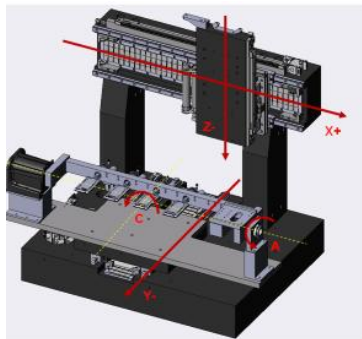


But how to manufacture
3D Printed Electronics?

Manufacturing of 3D printed electronics using a versatile platform strategy

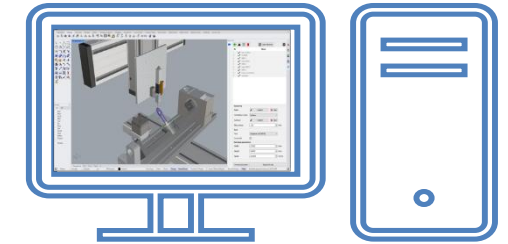
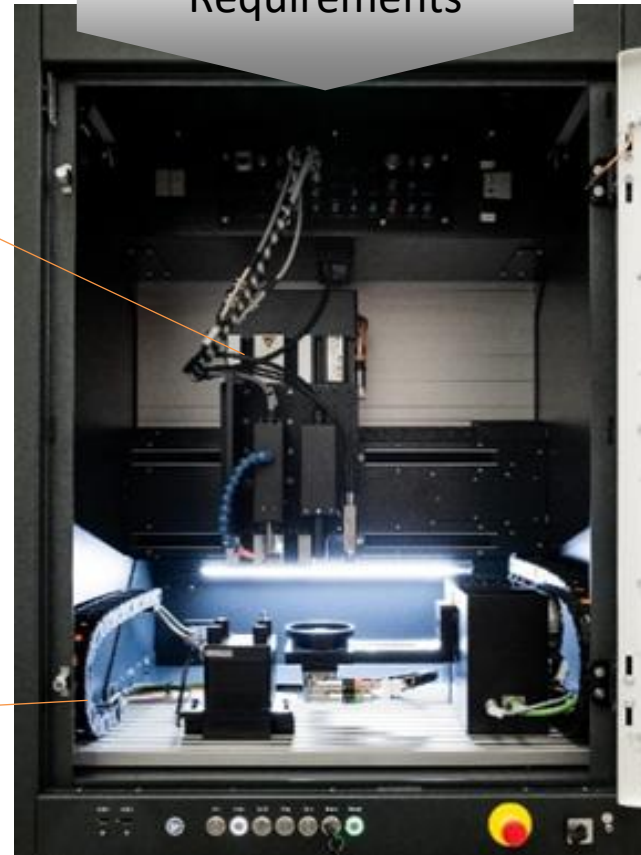


Manufacturing Tools and Processes



Hardware Platform

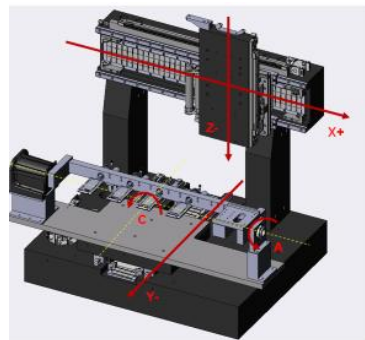
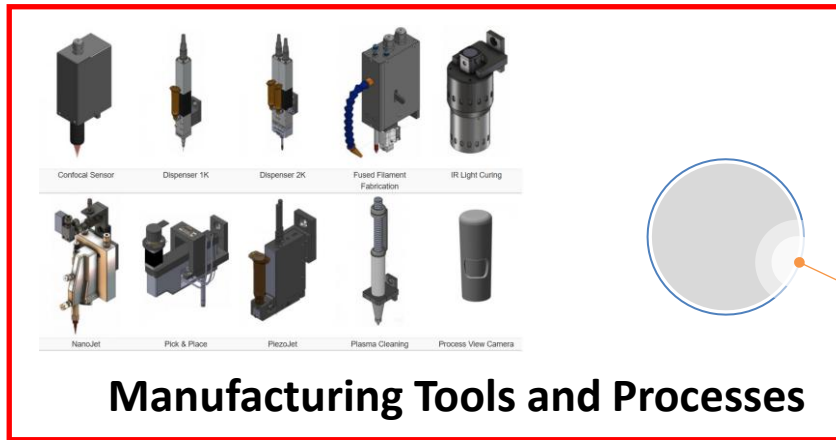
Application Requirements



Software Platform

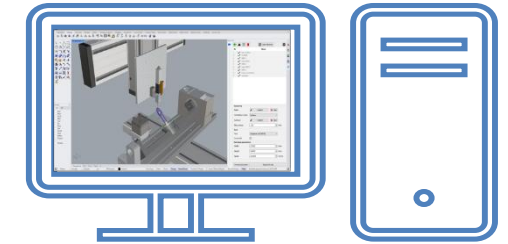
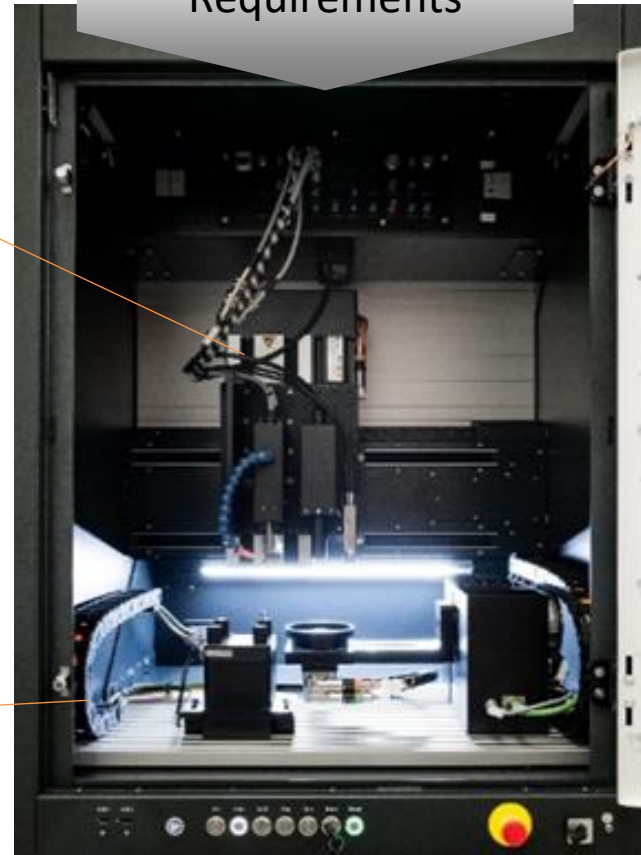


Manufacturing of 3D printed electronics using a versatile platform strategy



Hardware Platform

Application Requirements



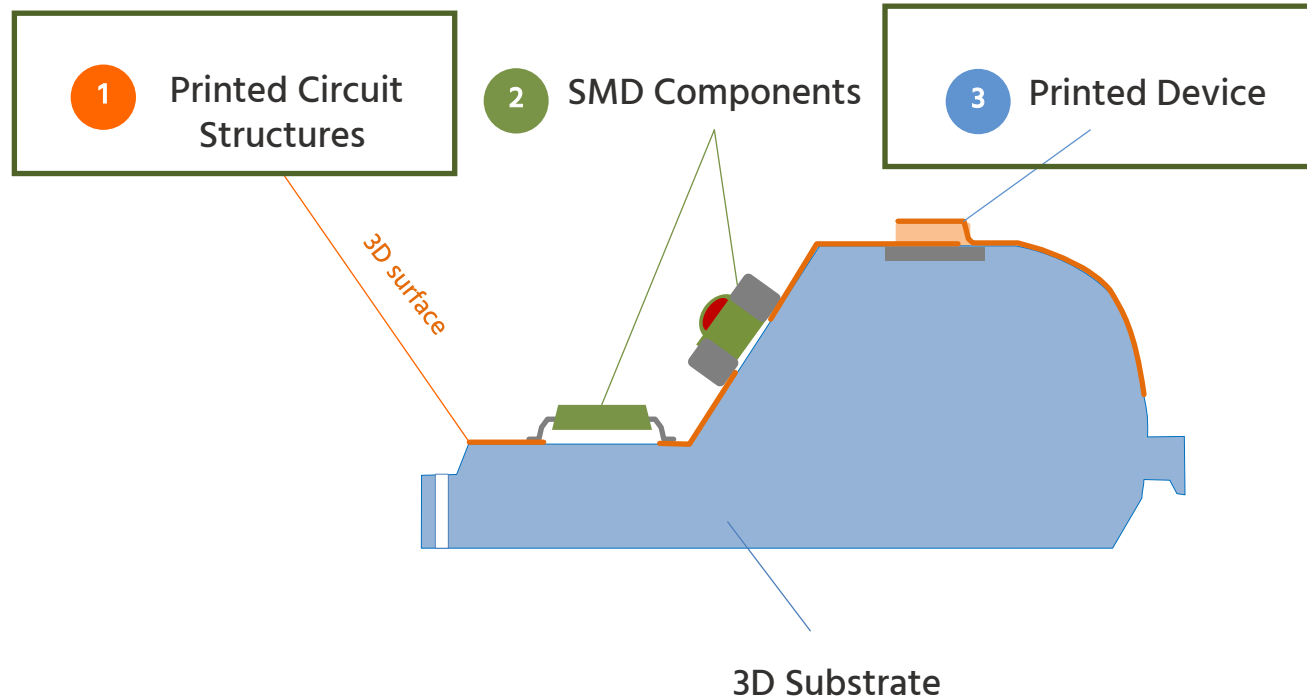
Software Platform



Main Features of 3D Printed Electronics



How to add electronic functionality to 3D shaped parts?



Method 1: Print on existing 3D parts (moulded, composite...)

Digital Printing Technologies

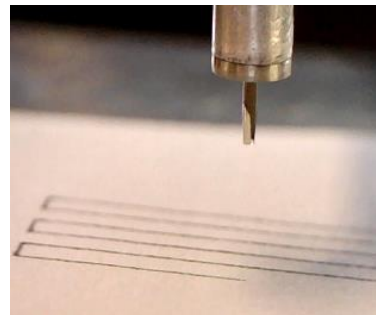
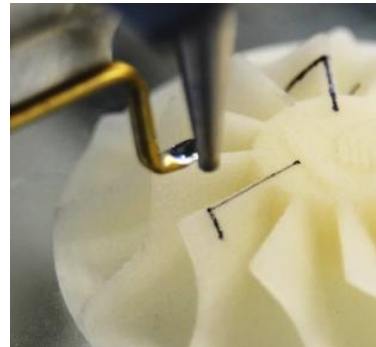


Print Processes commonly used to manufacture Printed Electronics:

Focus for manufacturing 3D Printed Electronics:

	Additive	Subtractive
Digital	• Inkjet printing • Extrusion • Laser induced forward transfer (LIFT) • Electrohydrodynamic (EHD) printing • Aerosol • Impulse printing • Print-then-plate • Laser direct structuring (LDS)	• Laser etching • Focused ion beam (FIB) etching IDTechEx Research
Analogue	• Screen printing • Flexography • Gravure printing • Offset printing • Reverse offset printing • Pad printing • Chemical and physical vapor deposition	• Photolithography (chemical etching) • Mechanical etching

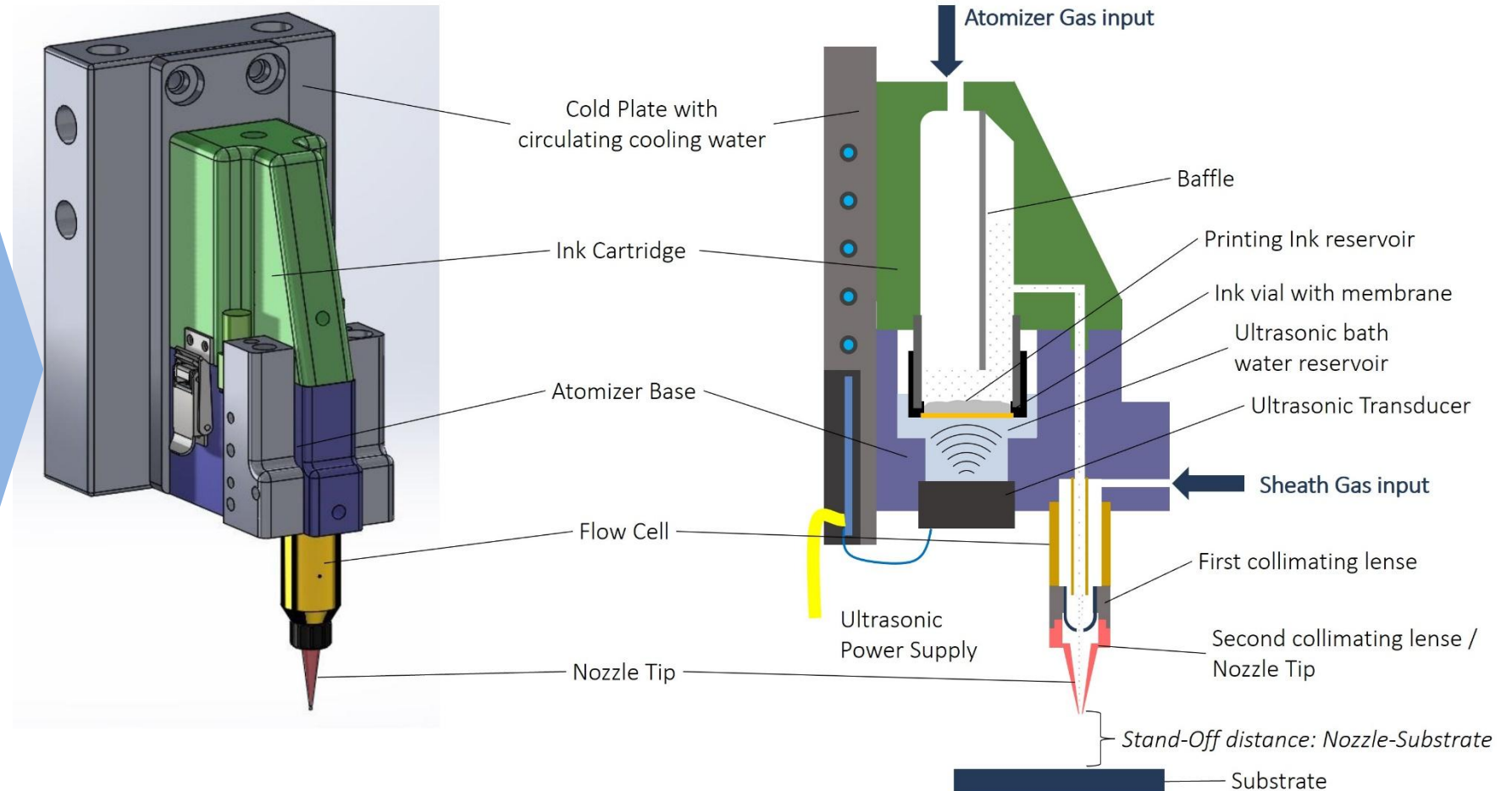
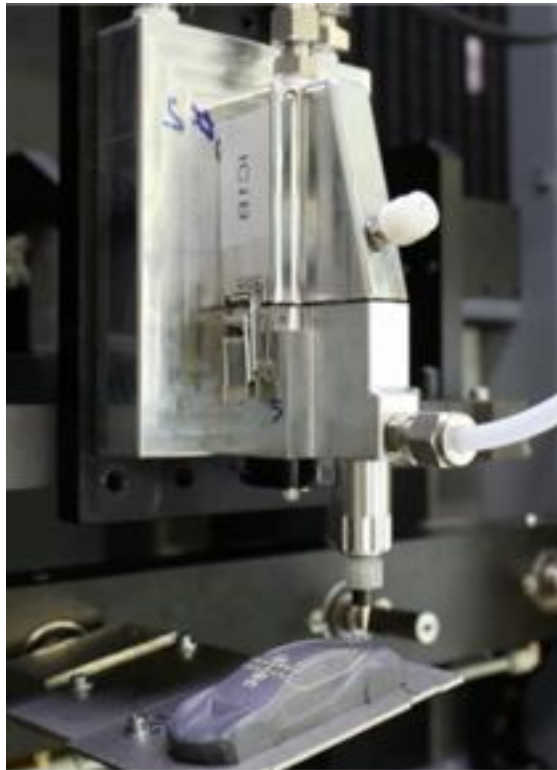
courtesy: PFFC-Online



Aerosol Based Printing using the Nanojet Technology



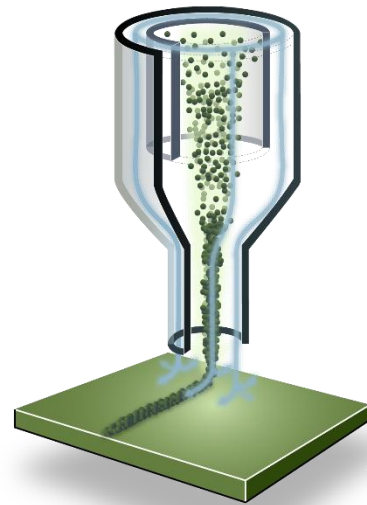
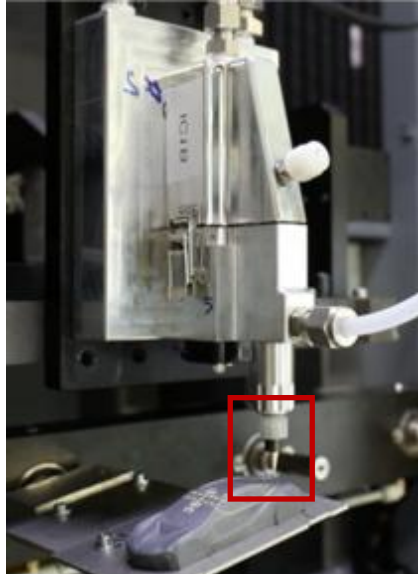
Contactless printing of fine line features:



Aerosol Based Printing using the Nanojet Technology



Focusing unit as key feature:



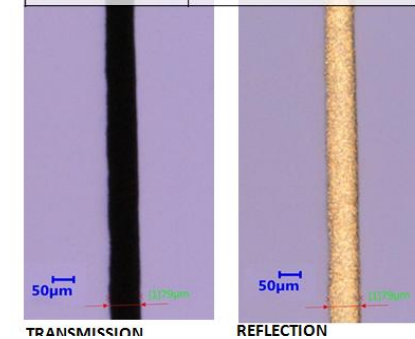
courtesy: Optomec

Characteristics:

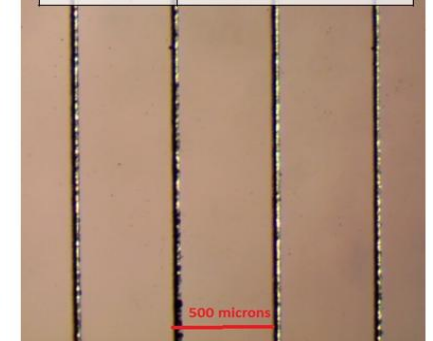
- low viscos inks (ca. 10 mPas), e.g. nanoparticle inks and pastes
- continuous atomization of functional ink → shutter required
- aerosol is aerodynamically focussed inside nozzle → stand-off of several millimetres
- Line width below 100µm achievable

PRINTED CONDUCTORS

INK	UTDots nanoparticle gold
SUBSTRATE	glass
PRINT SPEED	5 mm/s
LINE WIDTH	60 microns
# PASSES	2



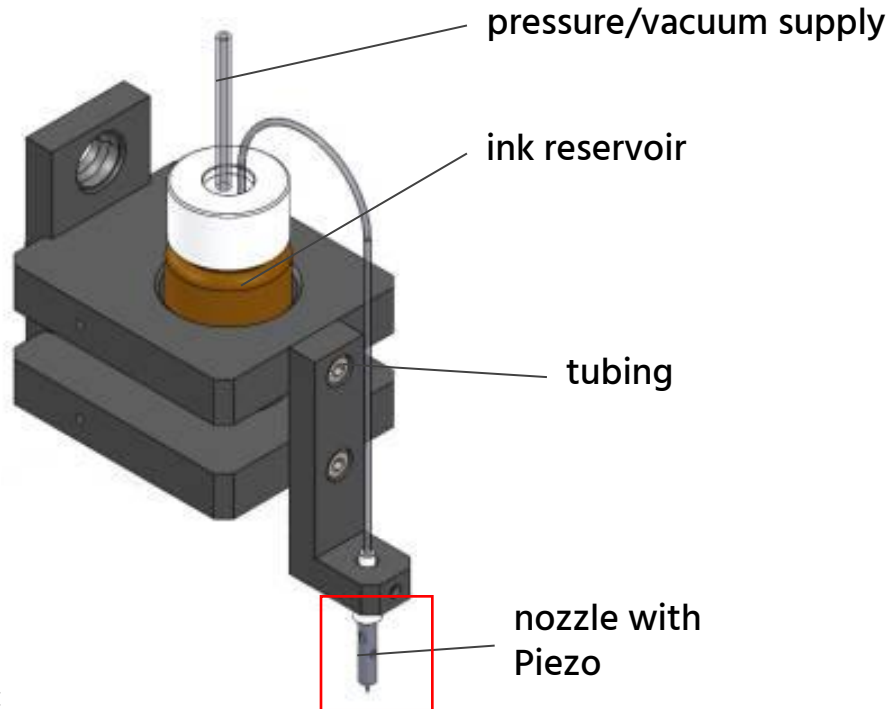
INK	UTDots nanoparticle silver
SUBSTRATE	glass
PRINT SPEED	5 mm/s
LINE WIDTH	40 microns
# PASSES	1



courtesy: IDS

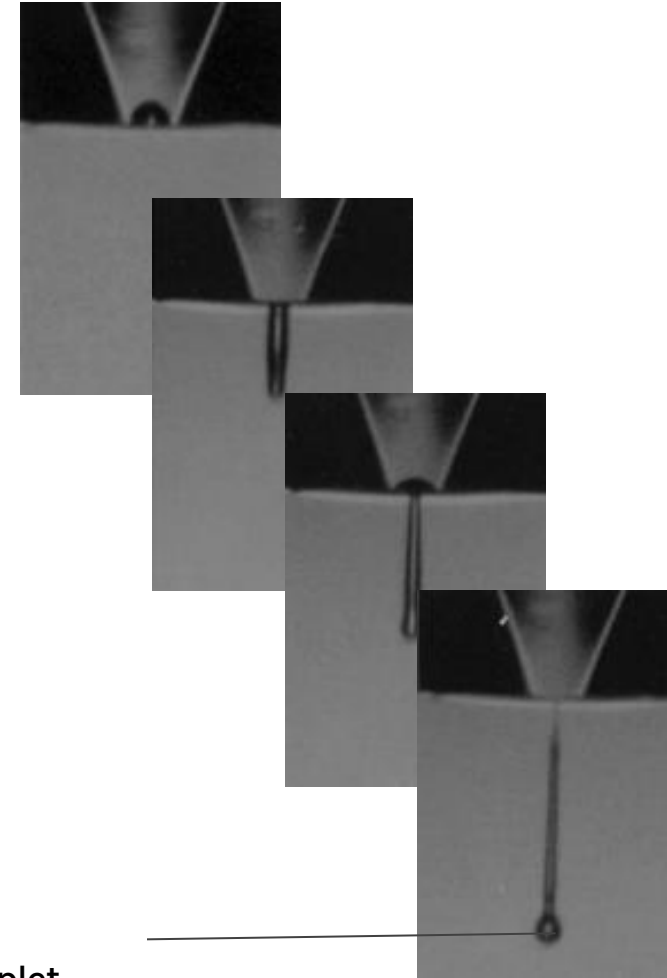
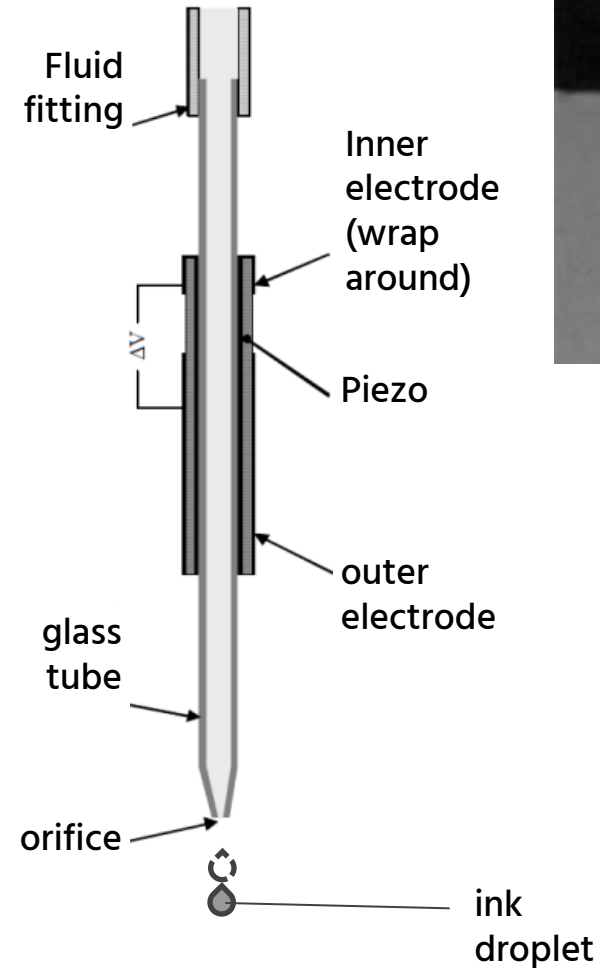
Single Nozzle Inkjet Printing

Contactless printing of fine line features:



Characteristics:

- low viscos inks (ca. 10 mPas), e.g. nanoparticle inks
- Drop-on-Demand process
- Short working distance limits 3D capabilities
- Line width below 100µm achievable

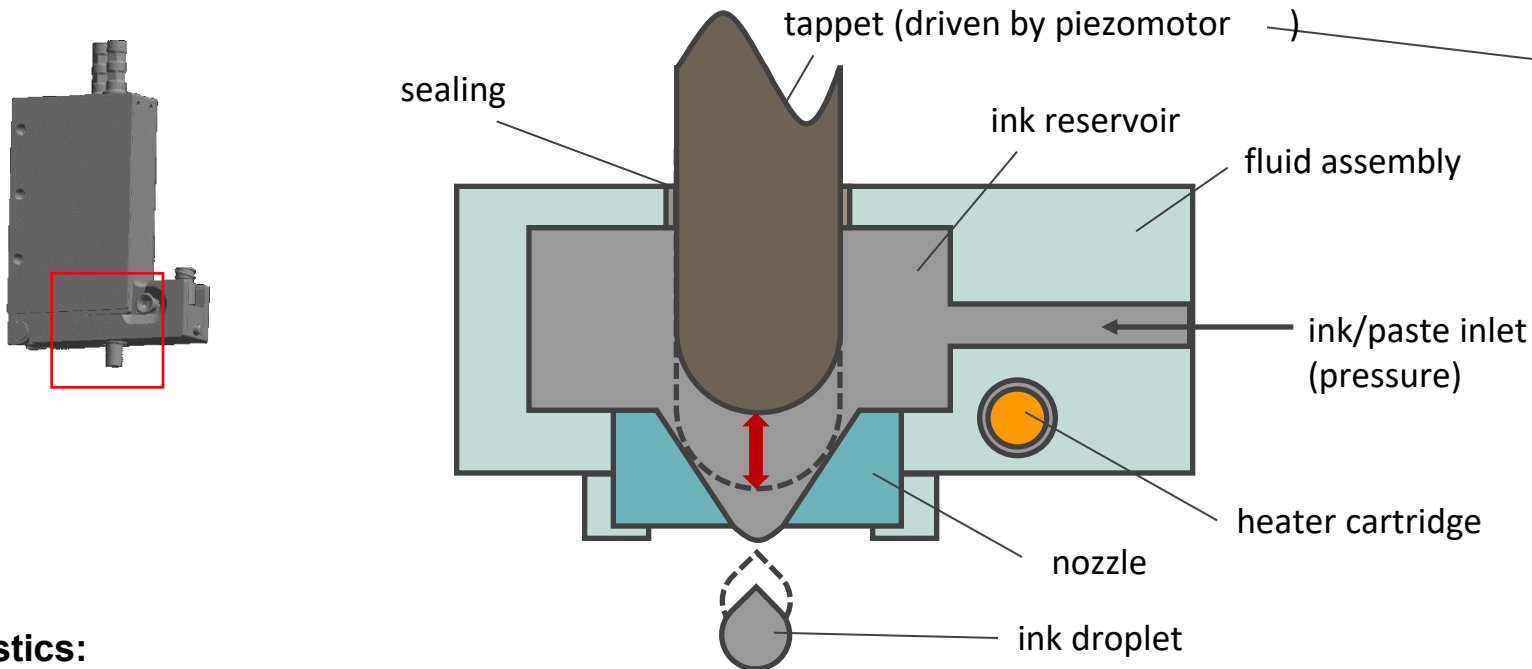


courtesy: MicroFab

Piezo Jet Printing

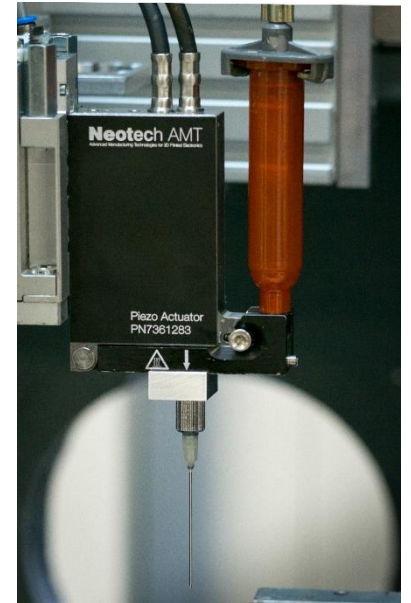
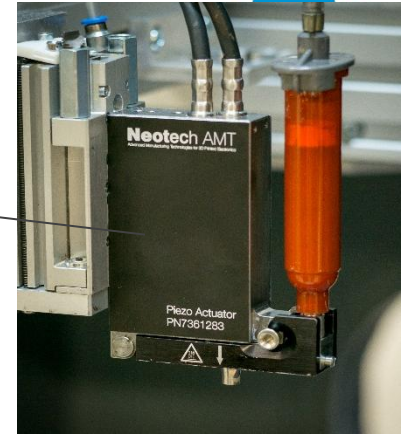


Jetting of functional materials like screen printing pastes:



Characteristics:

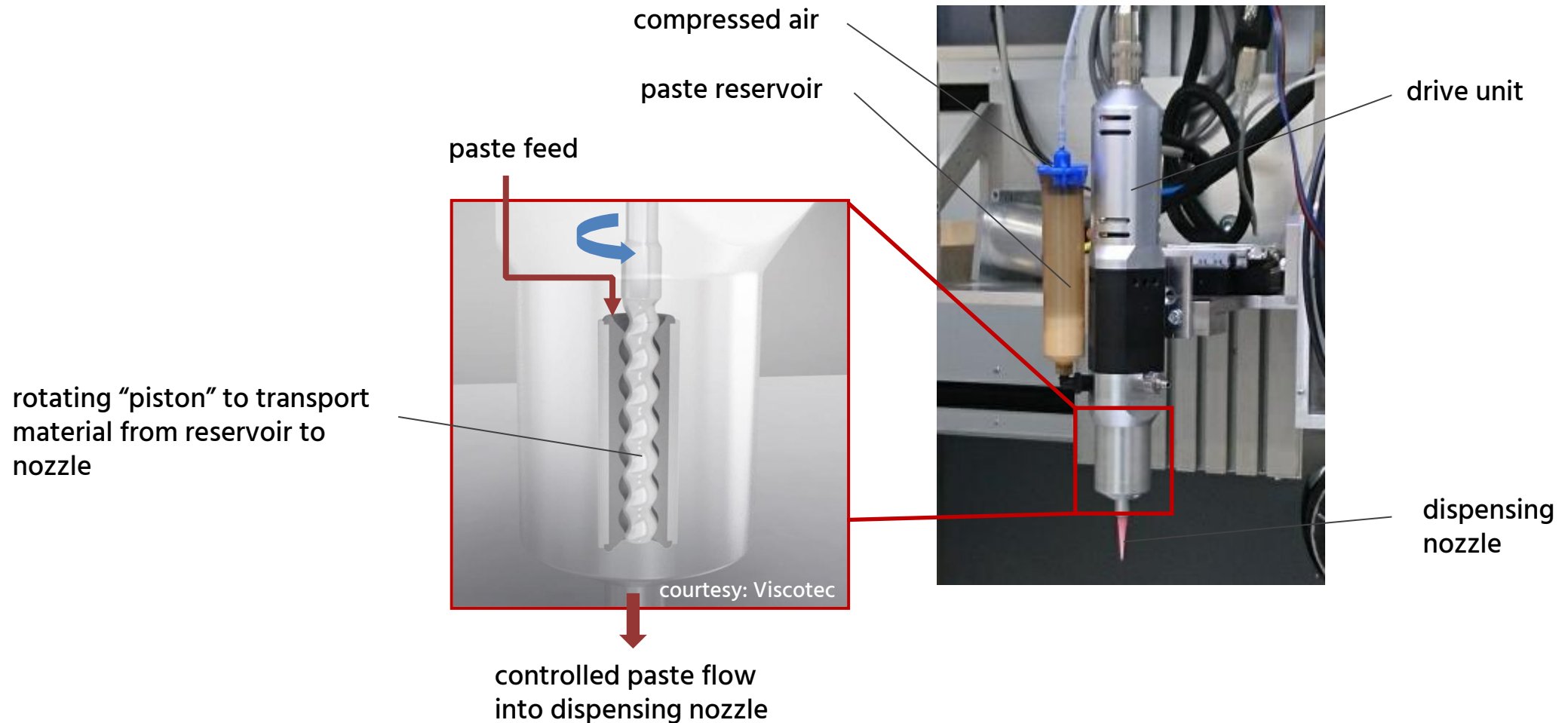
- Medium viscosity materials (1.000 – 20.000 mPas), e.g. nanoparticle pastes and adhesives designed for screen printing
- Drop-on-Demand principle
- Contactless printing with stand-off of approx. 1-10mm
→ Jetting process with variable stand-off
- Line width down to 200µm for particle free inks, 300µm for dispersions respectively



Needle Dispensing



Volume controlled Auger Dispensing according to endless piston principle:

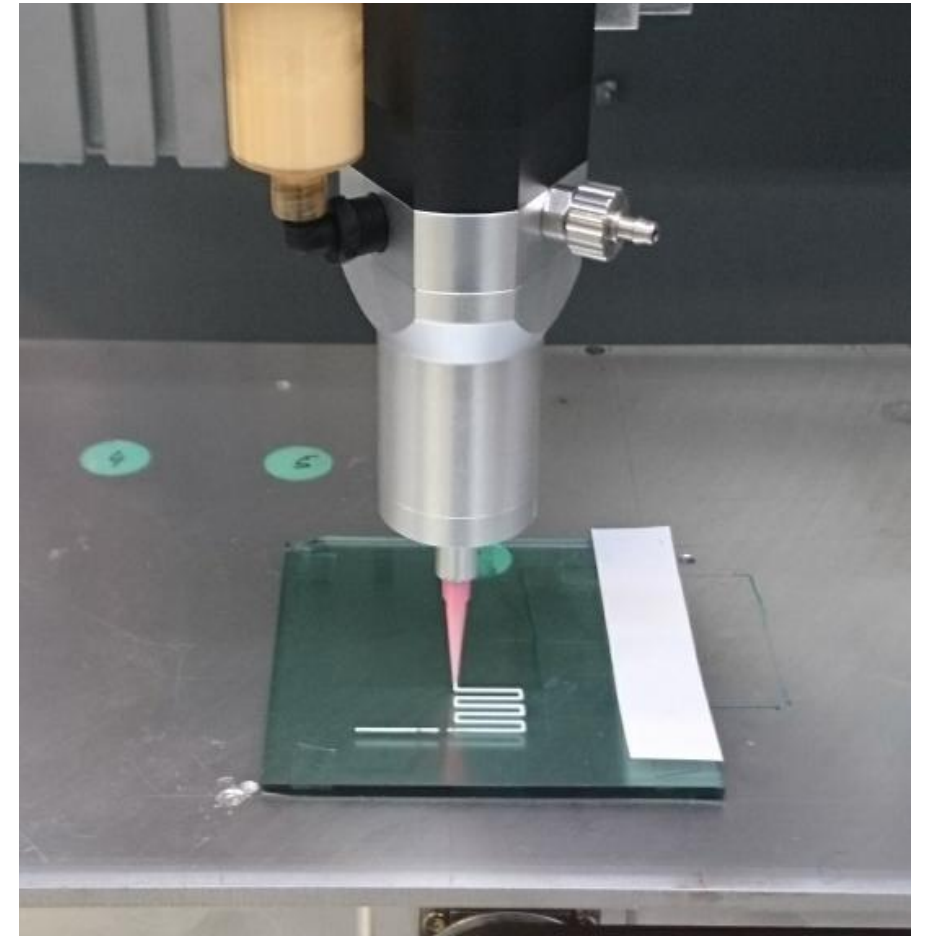


Needle Dispensing



Characteristics:

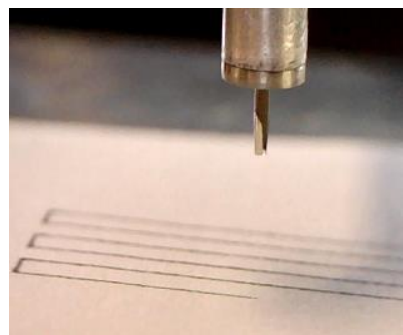
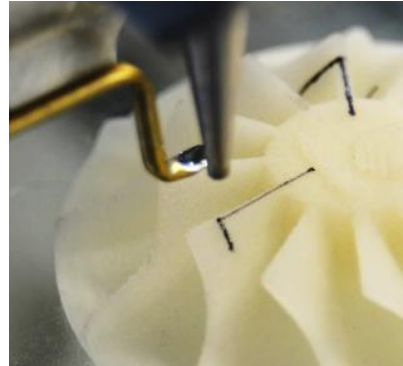
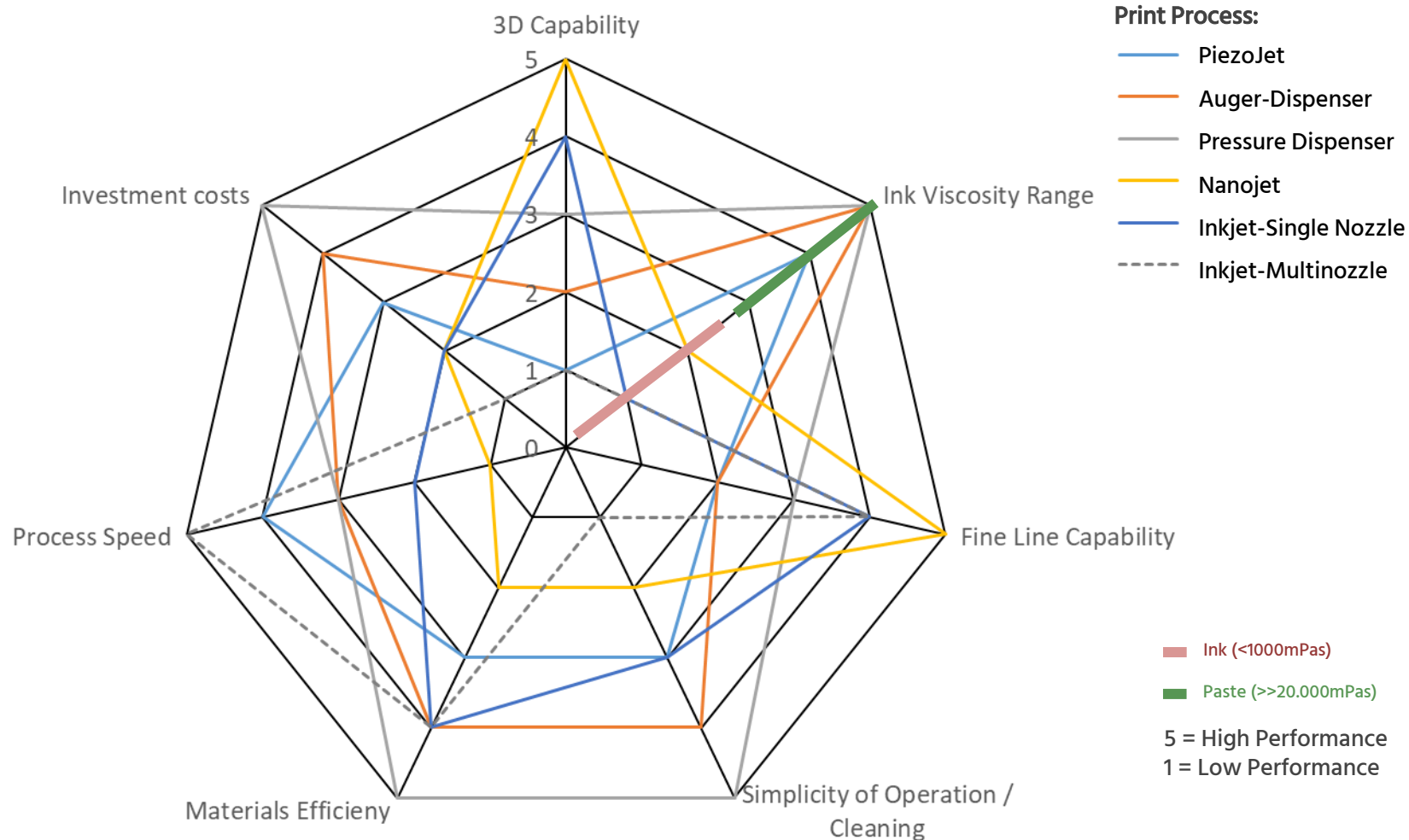
- High viscos inks and pastes (10.000 – 500.000 mPas)
- Continuous material extrusion; start/stop and output controlled via rotation speed and direction
- Stand-off between nozzle tip and substrate $\approx$ nozzle diameter
constant stand-off during printing influences print quality significantly – limited 3D capability
- Substrate height sensing required for
 - Fine line printing ($< 100\mu\text{m}$)
 - increased 3D capability



Comparison of different Print Technologies



Process selection driven by requirements resulting from the application and the functional material:



Options for Post-Treatment of Printed Materials



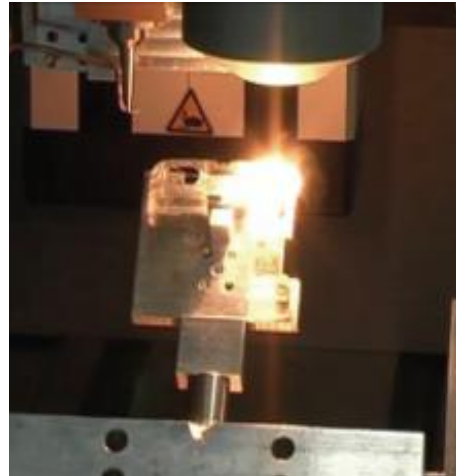
The deposited functional inks require post-processing to achieve their final electronical and mechanical performance:

Convection Oven:



- Off-line / separate station
- mature/reliable technology
- forced convection for 3D substrates
- wide field of application

Focused Halogen/Near-IR:



- In-line / system integration
- simple concept
- curing dependent on absorption
- spot-size of several millimetres (substrate is heated)
- broad wave-length spectra; combination with filters

Laser processing:



- In-line / system integration
- defined wave-length
- small spot size/high power density
- influence of absorption coefficient
- high safety requirements and integration effort (optical fibre)

UV-curing:



- In-line / system integration
- variable wave-length
- excellent controllability
- optical fibre limits options for integration; UV-LED heads under development;
- UV protection required

Embedding of Endless Wires



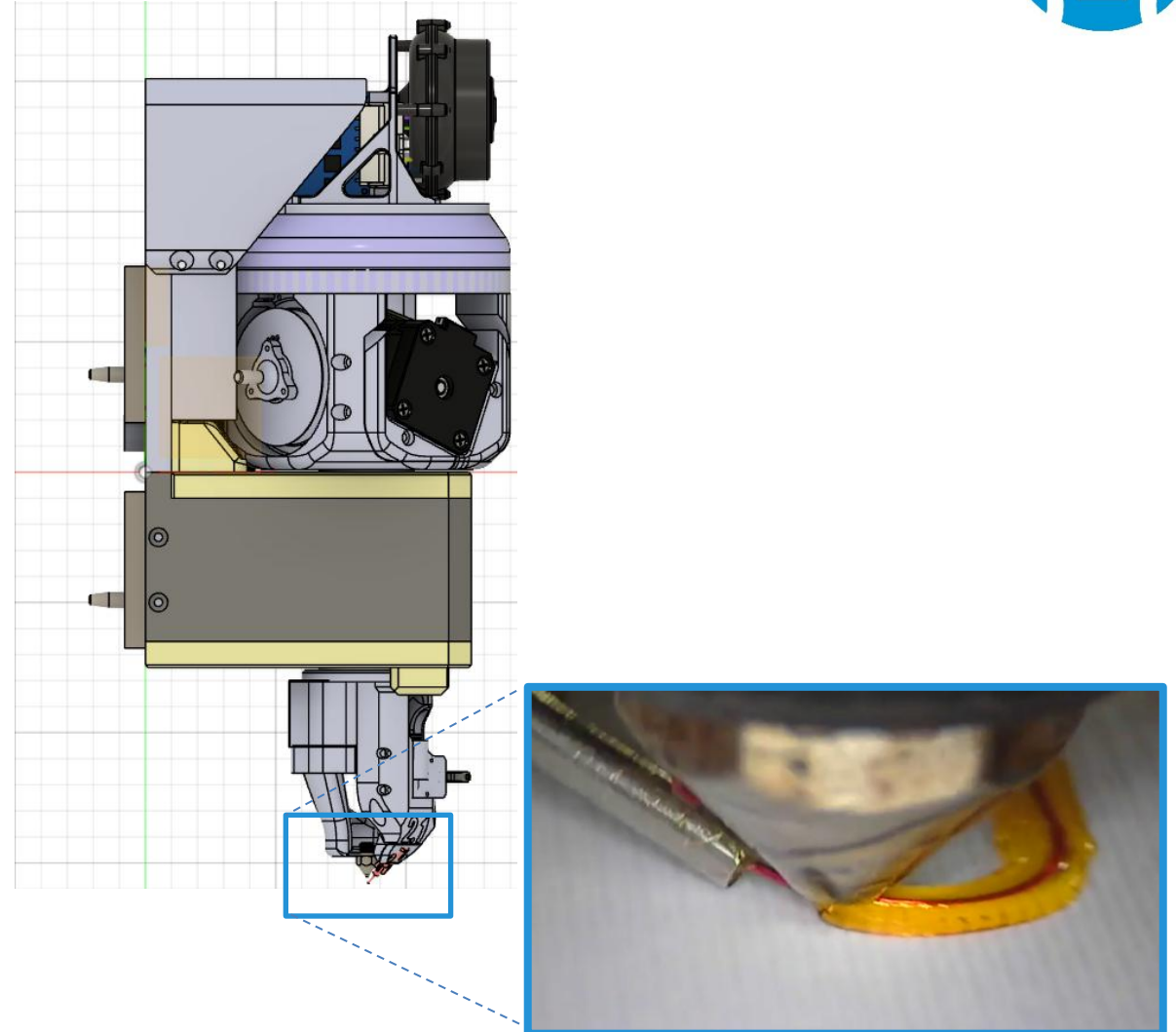
5-Axis Wire Embedding

Working Principle:

Precise extrusion of Cu-wires with simultaneous pinning process using Fused Filament Fabrication. Optional: winding process using endless wires or wire Start/Stop for complex circuit layouts with integrated cutting tool.

Advantages to printed conductive tracks:

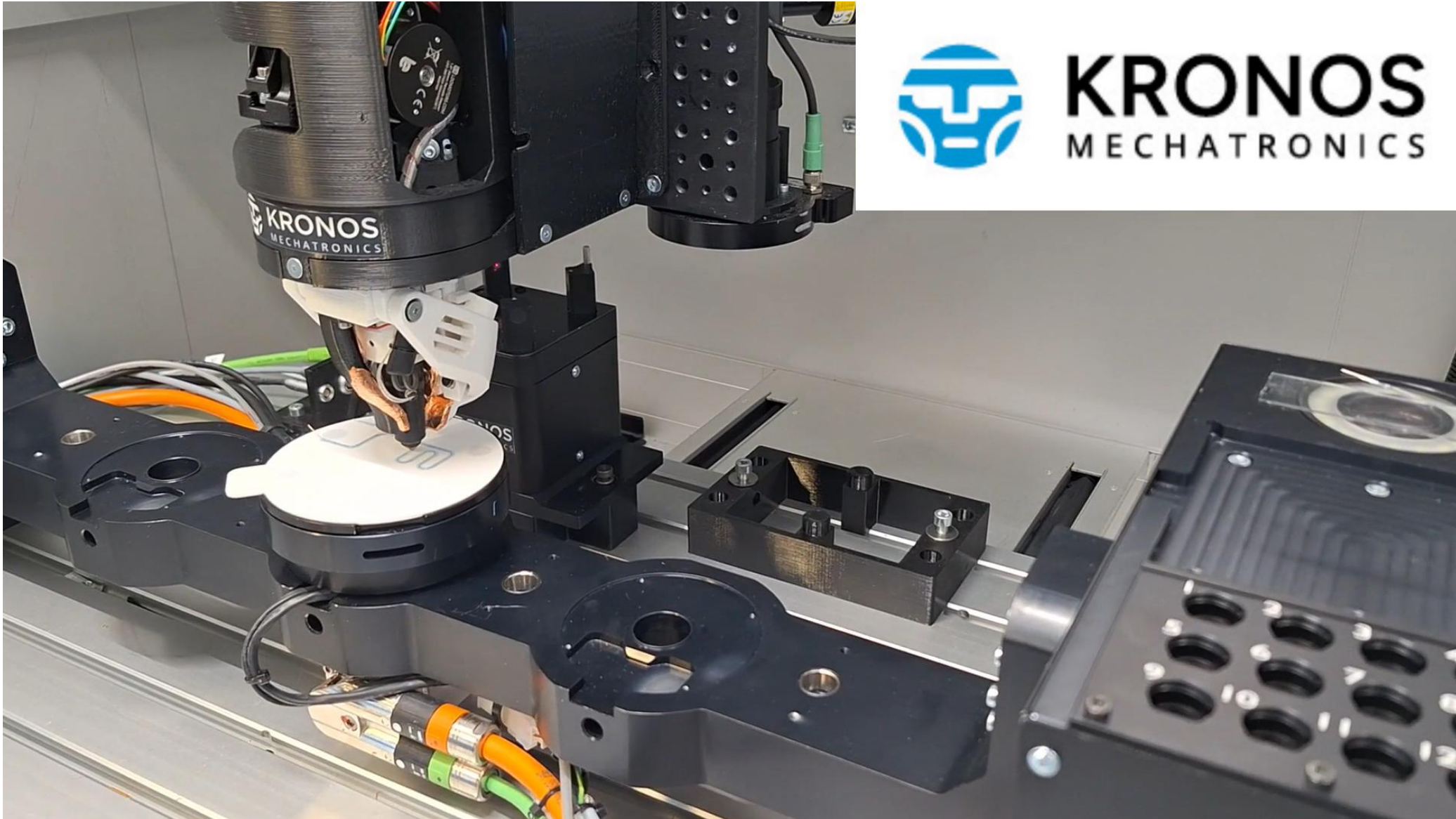
- > Robust and flexible conductors
- > Increased ampacity possible
- > No post-treatment of printed inks
- > Simplified cleaning processes compared to paste/ink handling
- > Various materials possible



Embedding of Endless Wires



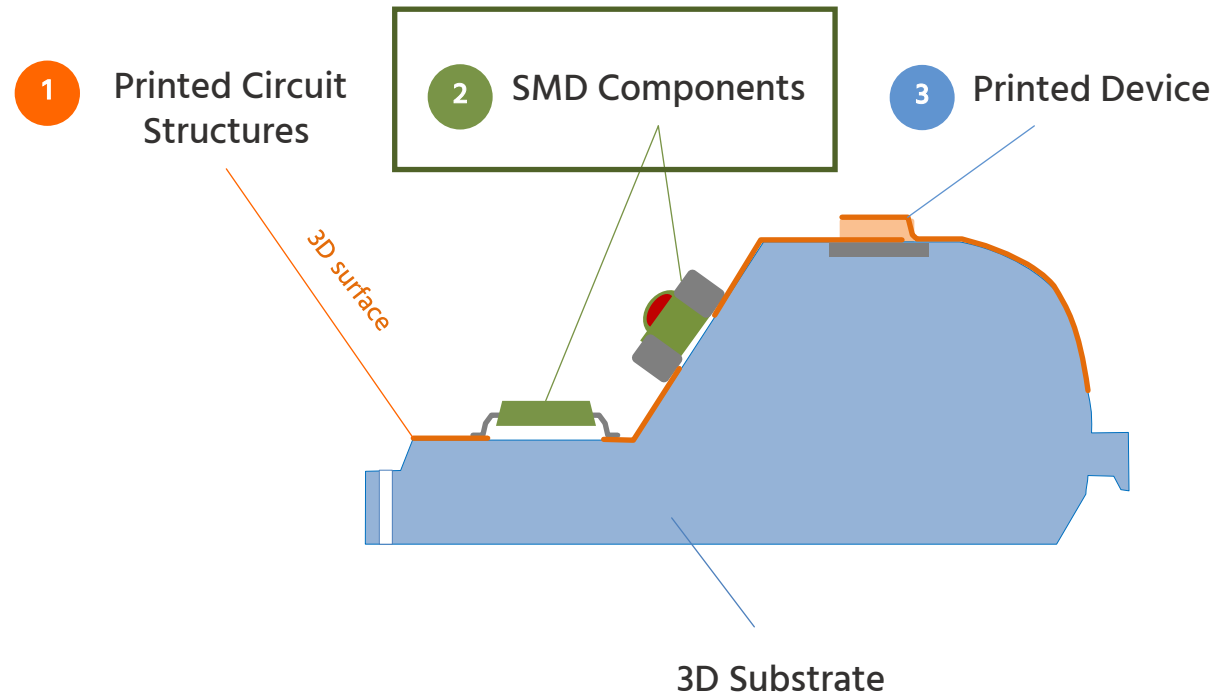
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Pick&Place Option for Component Insertion and Integration



How to add electronic functionality to 3D shaped parts?



Method 1: Print on existing 3D parts (moulded, composite...)

Electrical Interconnection of SMT Components

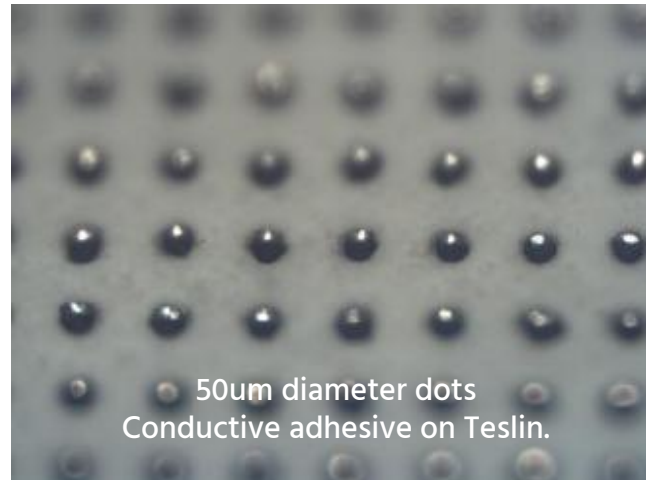


Image:
1-2: Courtesy Optomec Inc.
3: Courtesy Fraunhofer IKTS
4: Courtesy FAPS

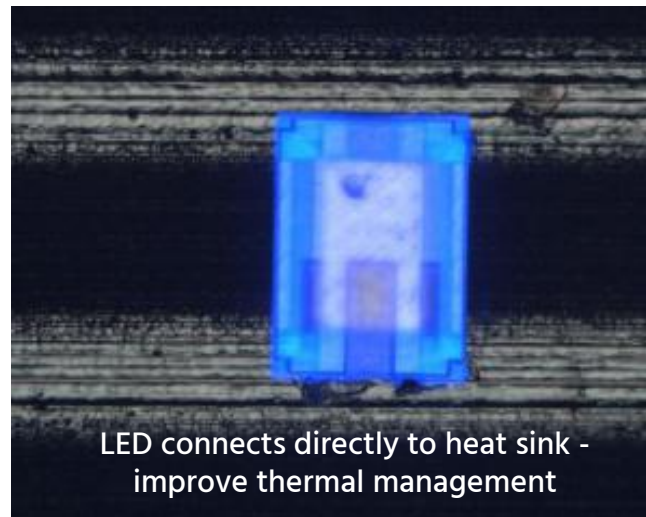
Soldering on printed conductive layers on 3D substrates:

- High temperatures required, usually low thermal stability of substrates
- Standard solder materials “dissolve” thin silver layers (metallurgy)
- Application of small solder depots on 3D surfaces is challenging

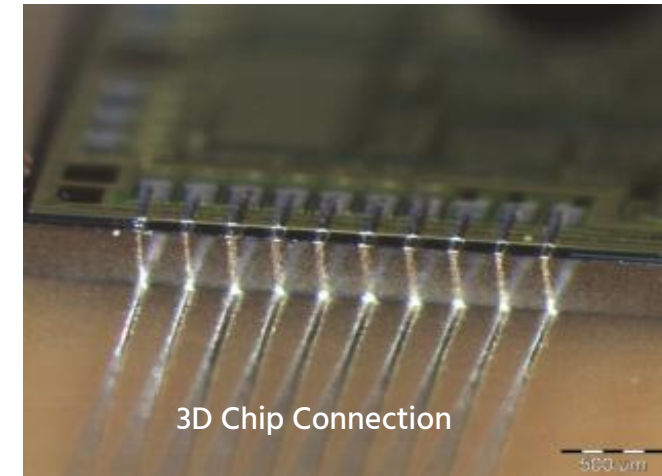
1. Print Conductive Adhesive – Flip Chip



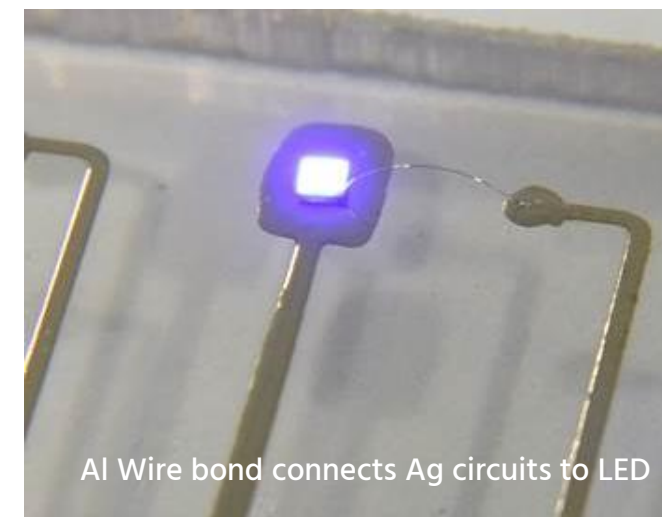
2. Use Circuit material to fix SMD



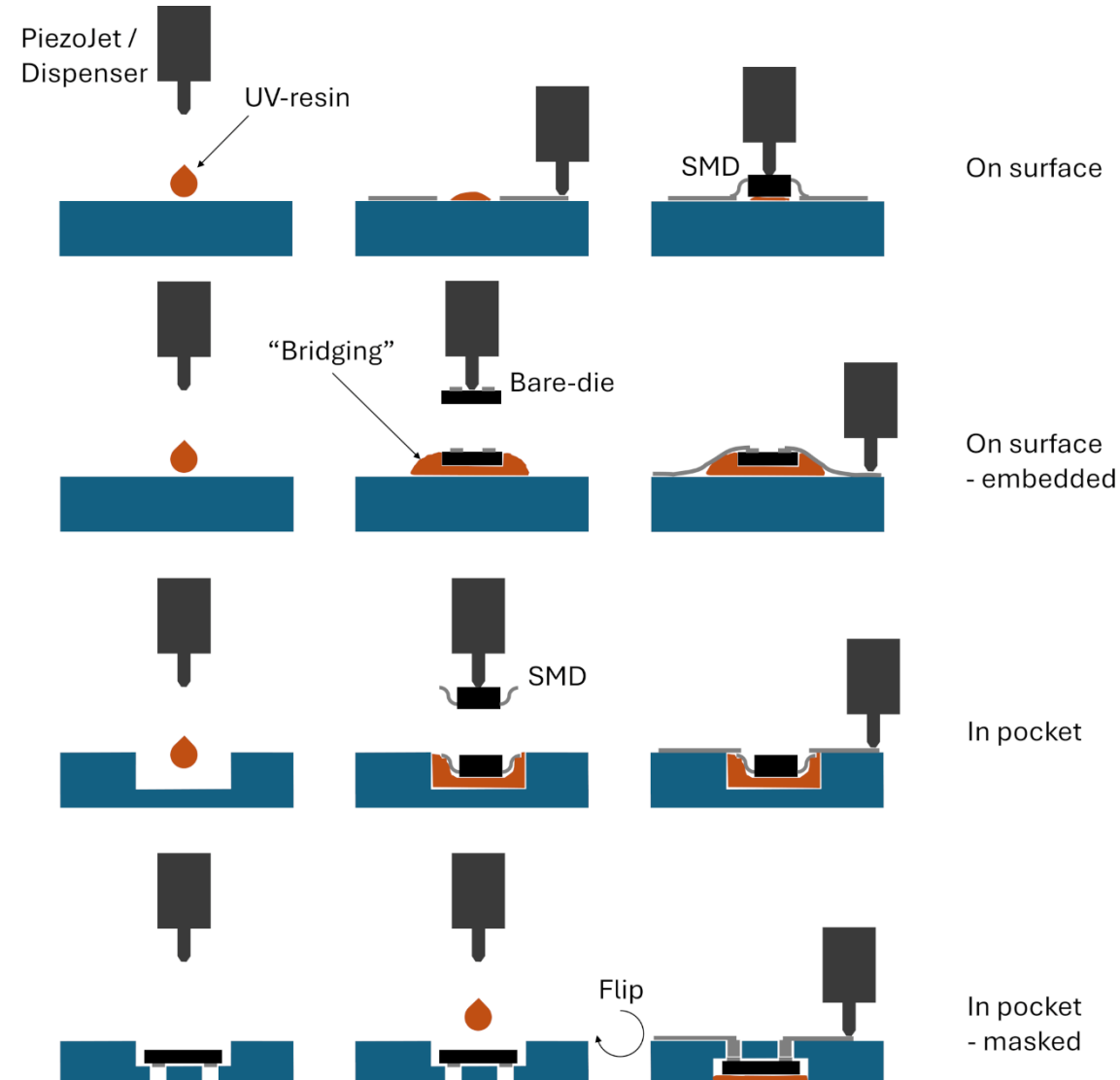
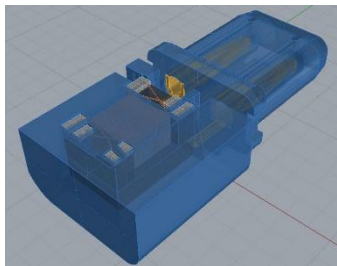
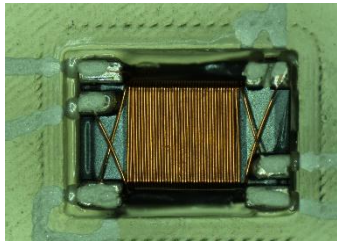
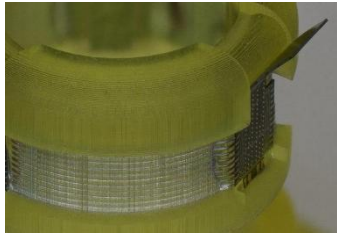
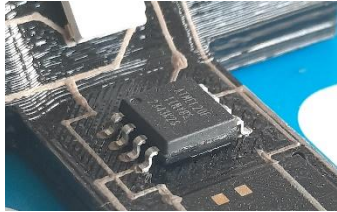
3. Directly Write Interconnects



4. Wire Bond onto printed circuit



Basic options for the integration of electronic components



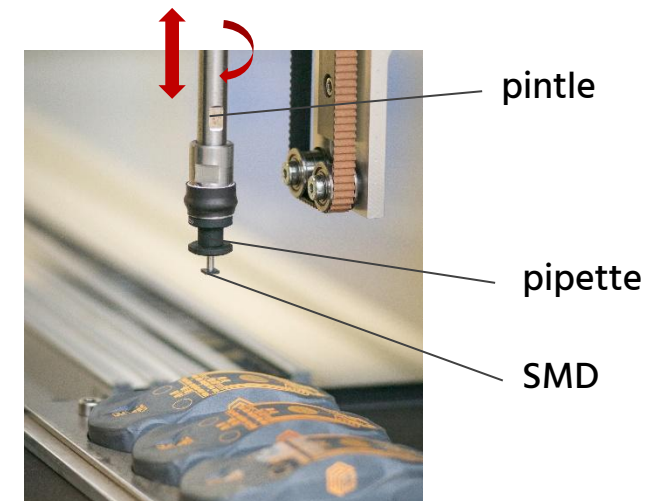
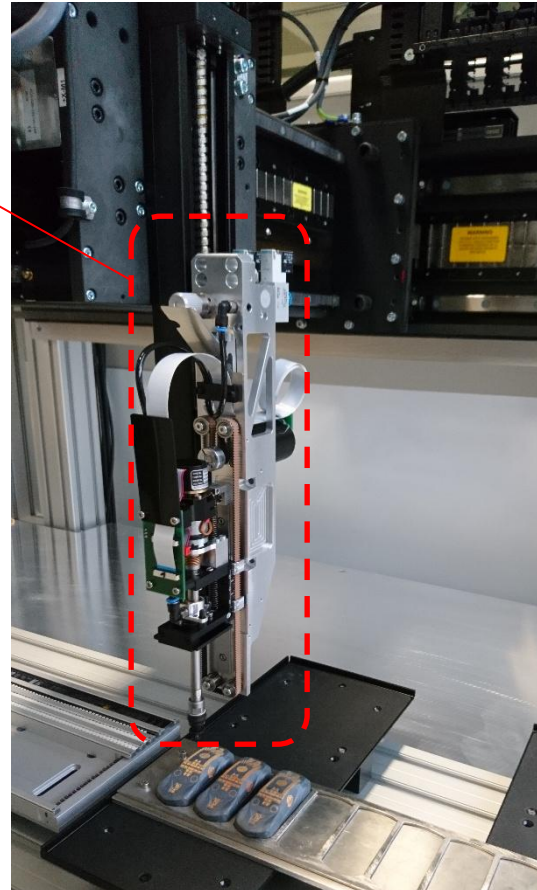
Pick&Place for Component Insertion and Integration



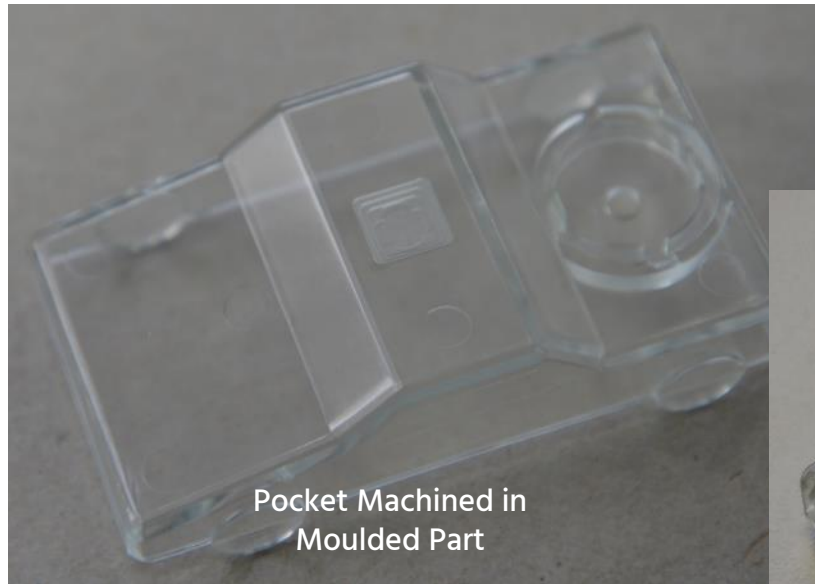
Positioning of SMD components in and on 3D substrates:

Integration of Pick & Place module in 5-axes platform:

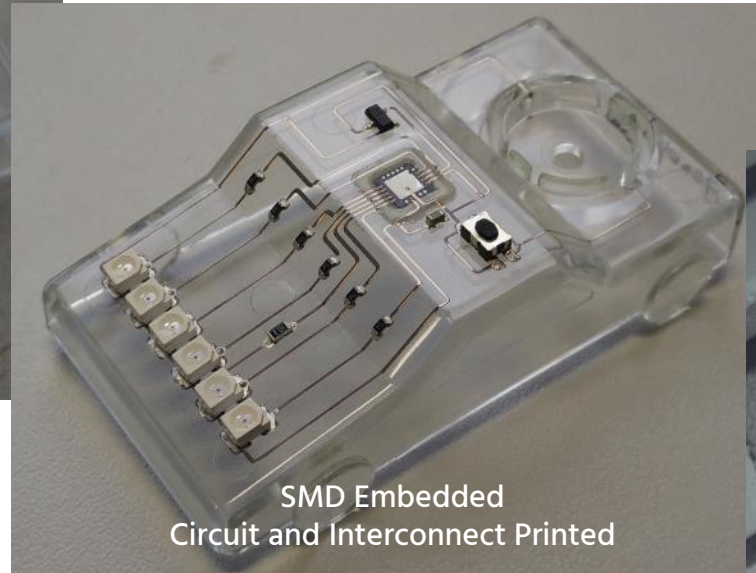
- Additional z-axis and rotation unit
- Interchangeable vacuum pipettes according to component size
- SMD component supply using standard feeders and trays
- Optional component camera for precise component placement



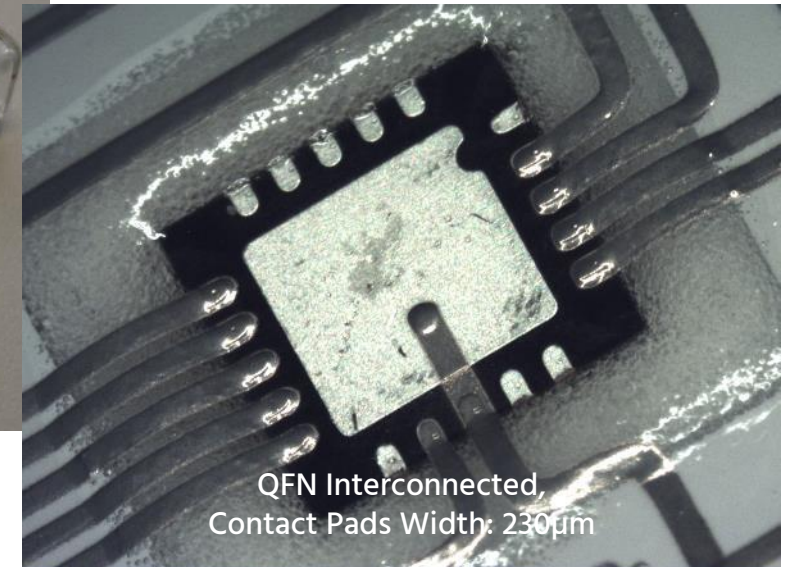
Printed Interconnects for Flipped and Embedded SMDs



Pocket Machined in
Moulded Part



SMD Embedded
Circuit and Interconnect Printed



QFN Interconnected,
Contact Pads Width: 230µm

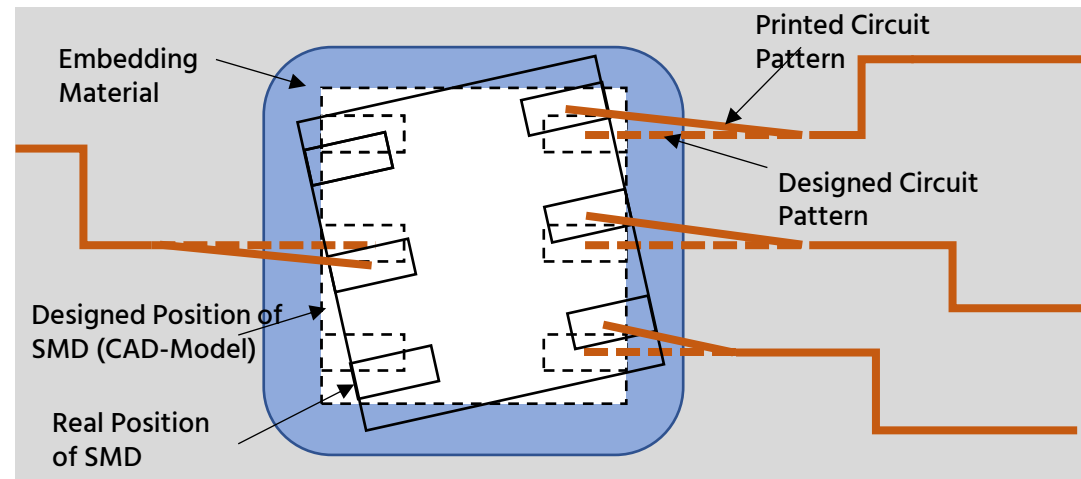
Benefits:

- No application of small volumes of interconnection material (ICA, Solder etc.)
- Single process step for printing and interconnection
- Low temperature processing (depending on ink)
- Processing of bare dies, mechanical fixing and protection of the component

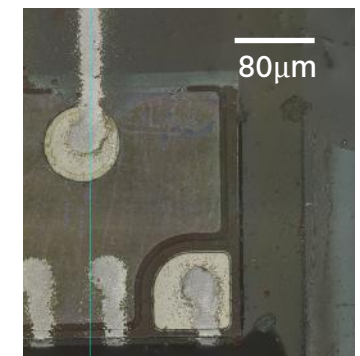
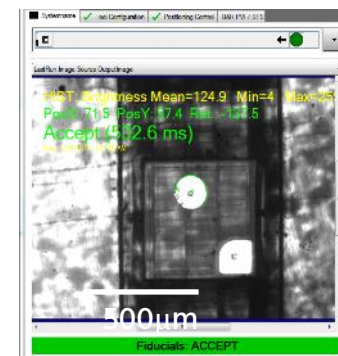
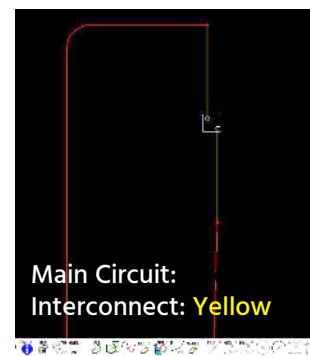
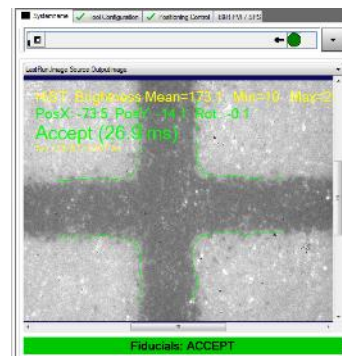
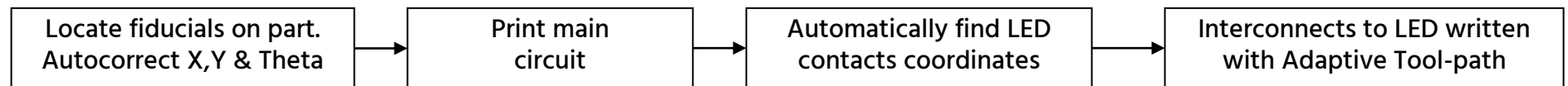
Printed Interconnects for Flipped and Embedded SMDs



Displacement due to floating of the component during placement process into liquid/adhesive:



Solution:
Adaptive toolpath:

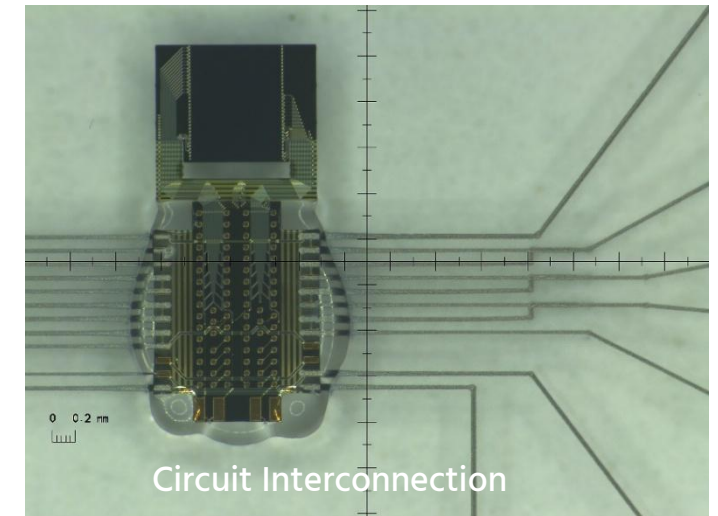


Printed Interconnects for Flipped and Embedded SMDs



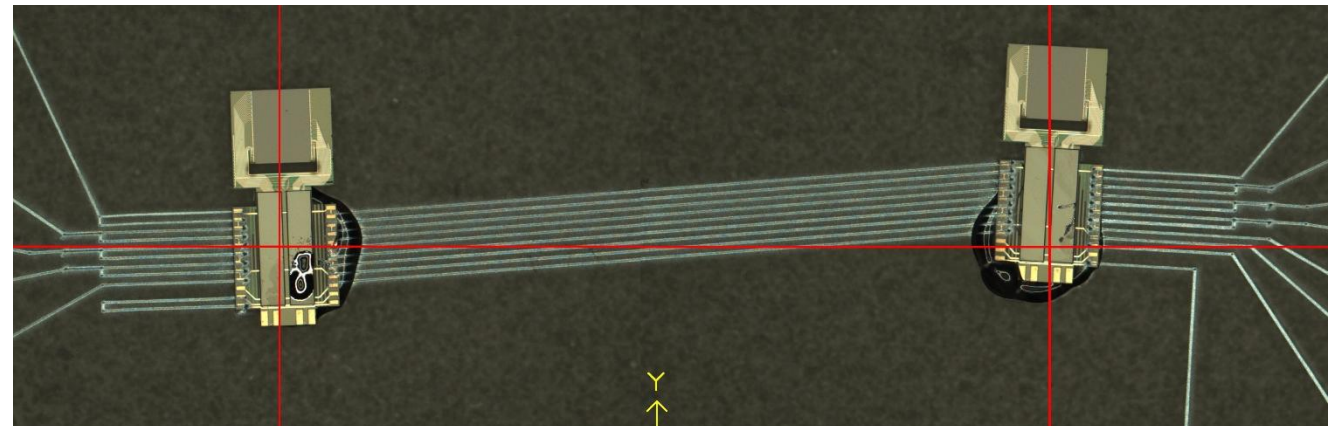
Initial P&P of CMUT-ASIC array in 2D

Interconnects are Aerosol Printed at ca. 40um line width.



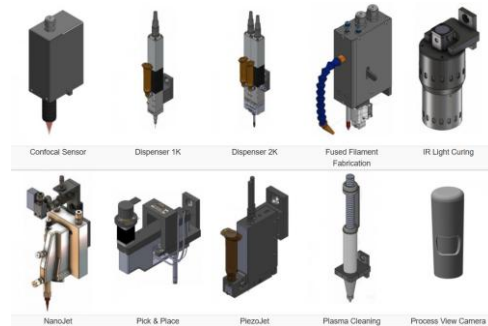
Placement of multiple ASICs in UV curable adhesive
→ component displacement during P&P.

Adaptive toolpath strategy compensates displacement and
adjusts toolpaths for printing interconnect traces.

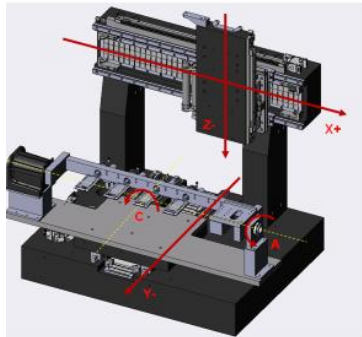


AMPERE

Manufacturing of 3D printed electronics using a versatile platform strategy

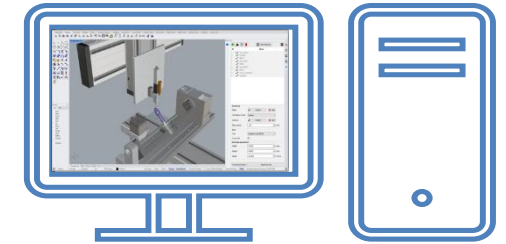
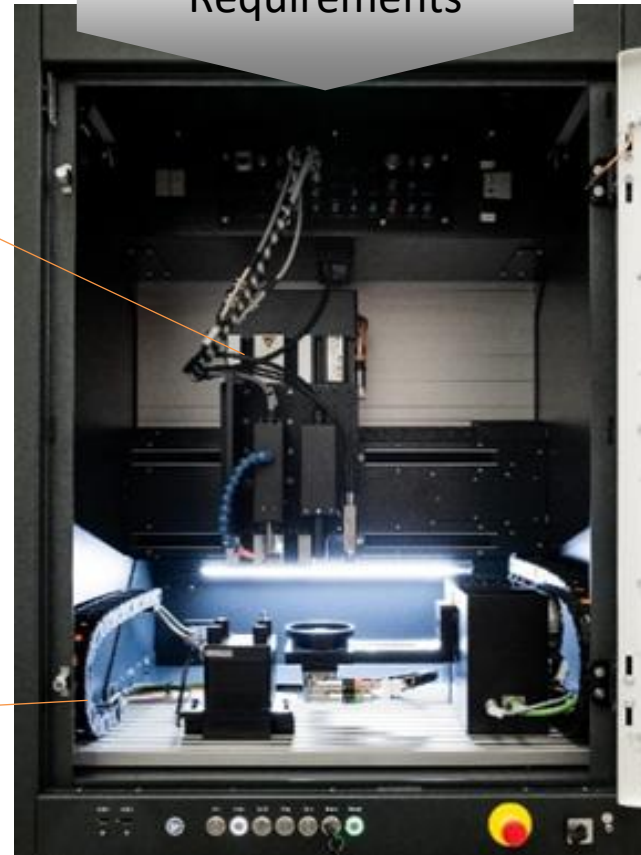


Manufacturing Tools and Processes



Hardware Platform

Application Requirements



Software Platform

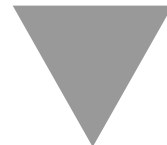


Motion Systems / Print Platforms



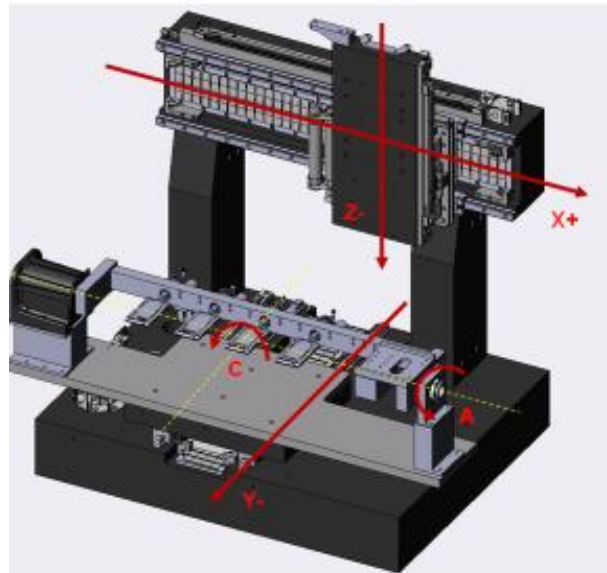
Selection of a suitable CNC multi-axis system:

Environmental Requirements and Influences



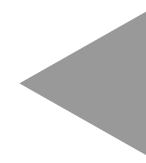
Print Technology

- Mass Output of Printhead
- Standoff Nozzle to Substrate
- Design and Complexity of Head
- Variation of Print Direction



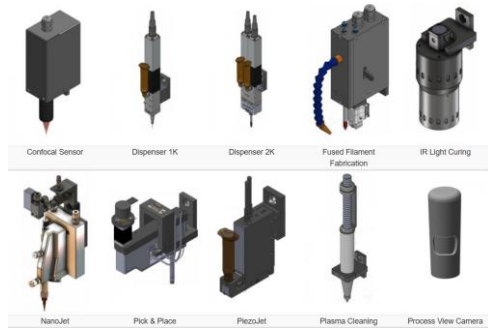
Part/Substrate

- Parts dimensions
- Weight and Material
- Required accuracy for printed pattern (overall, local)
- Shape and Accessibility

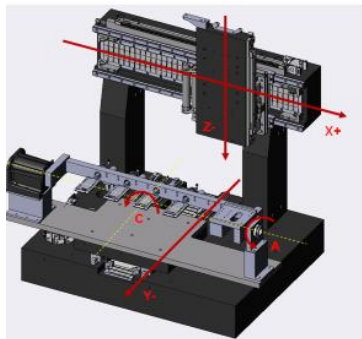


- Axes Configuration and Performance
- Performance and Capabilities PLC
- Design and Stiffness of Frame

Manufacturing of 3D printed electronics using a versatile platform strategy

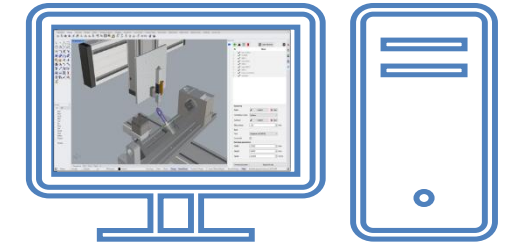
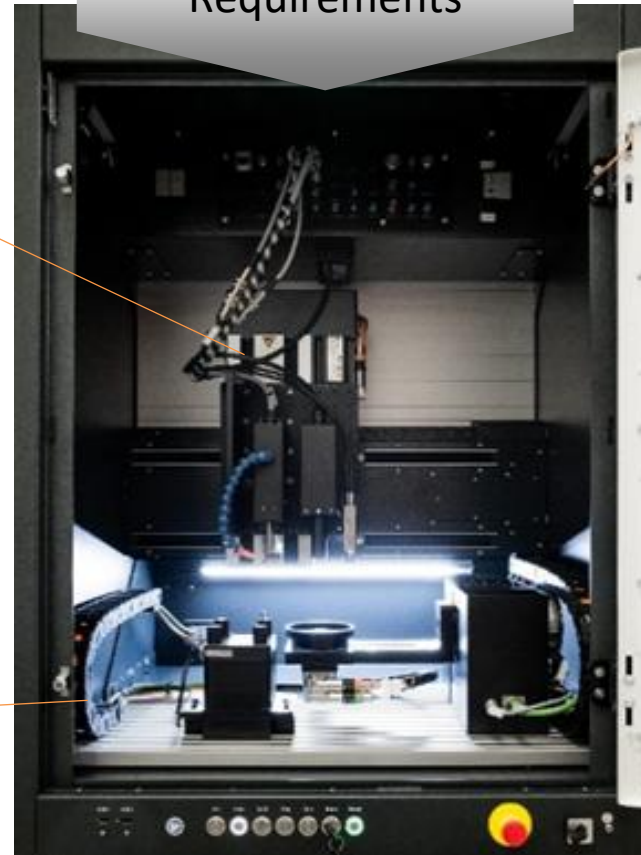


Manufacturing Tools and Processes



Hardware Platform

Application Requirements



Software Platform



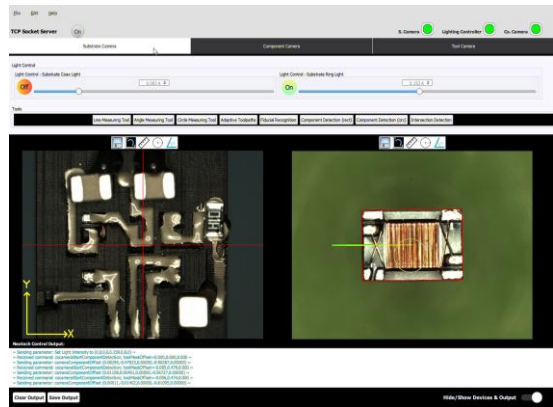
Software Framework for Automated Production



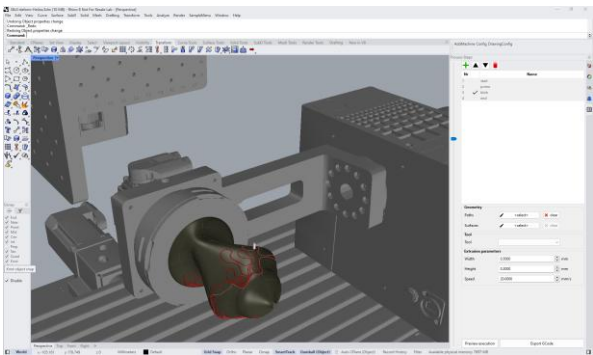
Programmable Logic Controller



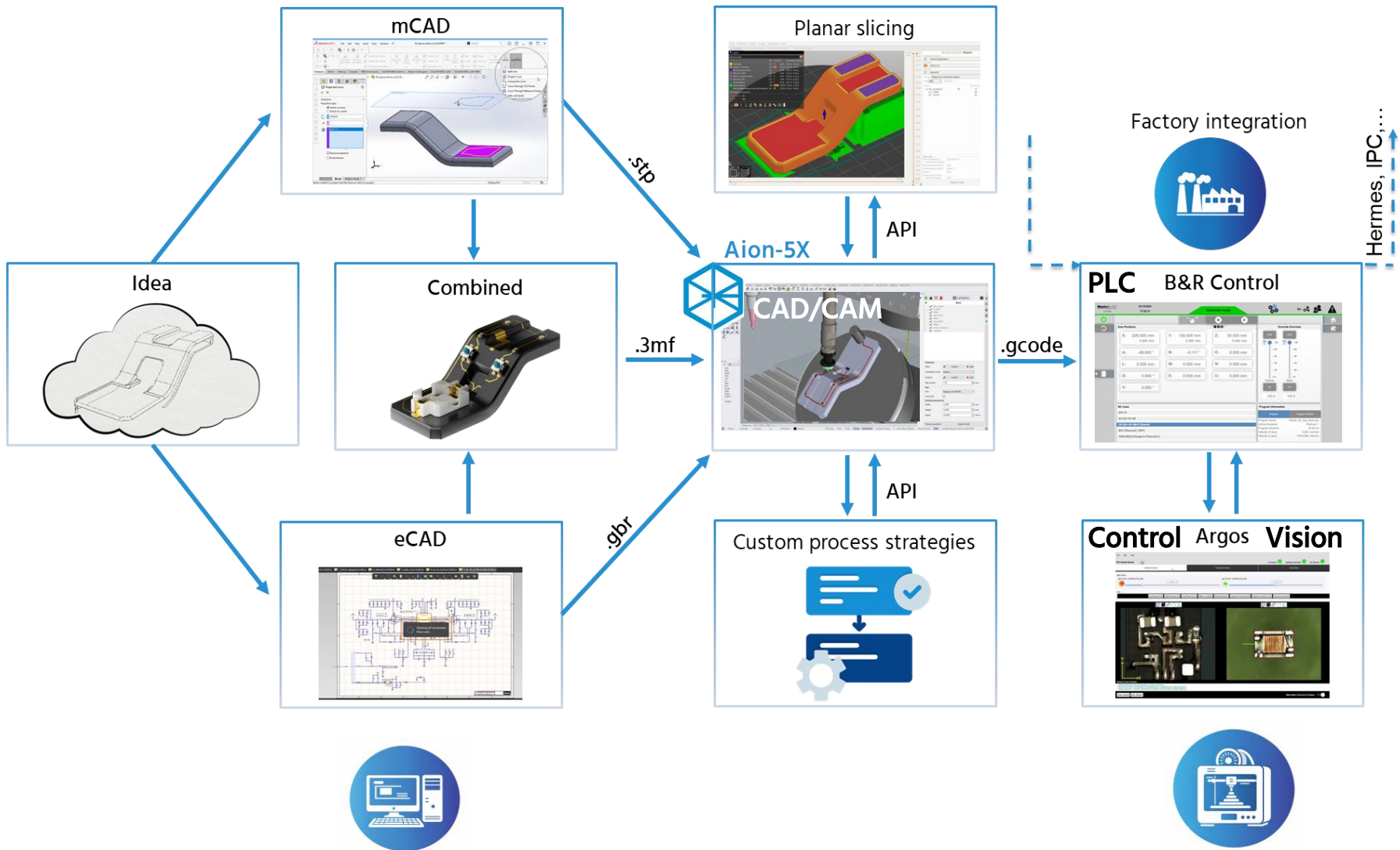
Control and Vision System



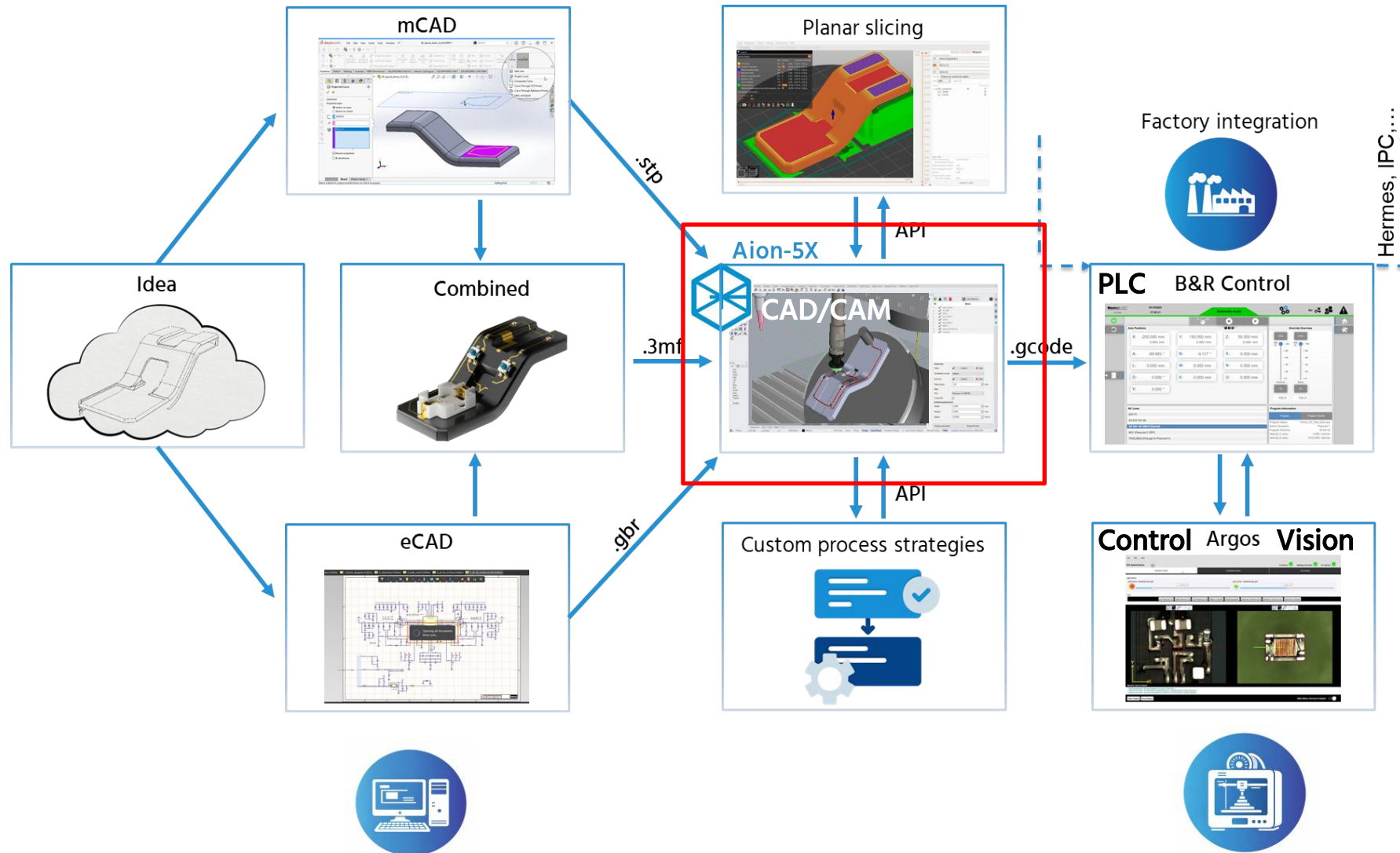
CAD/CAM Software



Software Ecosystem & Interoperability



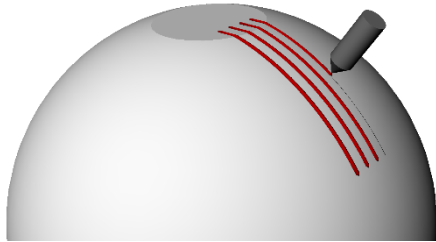
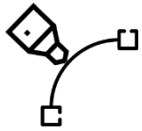
Software Ecosystem & Interoperability



CAD/CAM - Process Strategies (Aion 5X)

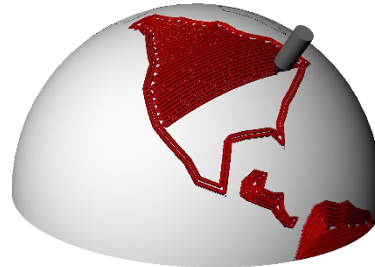


Aion-5X enables multiple, geometry-driven printing strategies from a single software platform – adapting each process to the part, not the other way around.



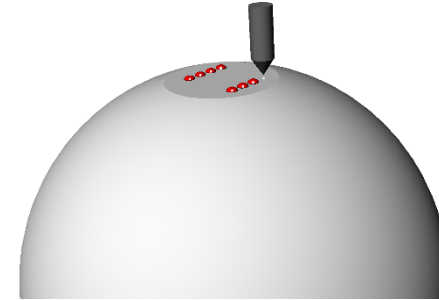
Contour

Lines & wires



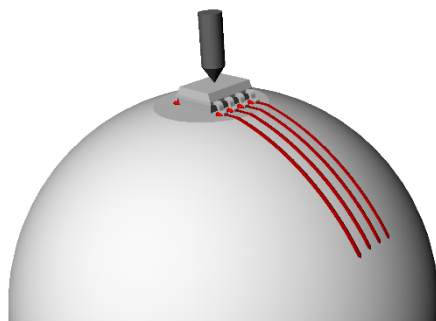
Surface

Freeform surface coating



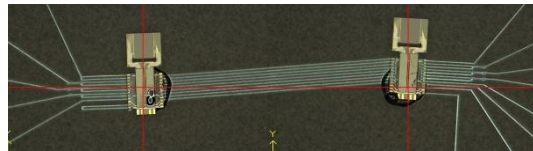
Point

Mount & glue



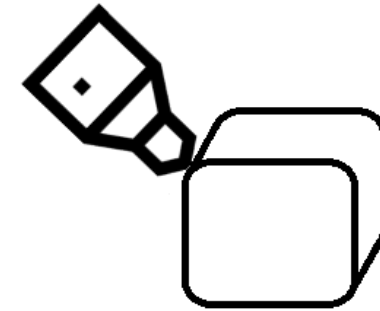
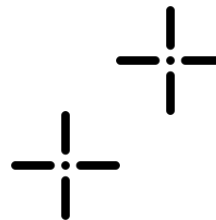
Pick & Place

SMD components



2D Alignment

Adaptive positioning



Volume

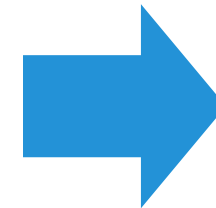
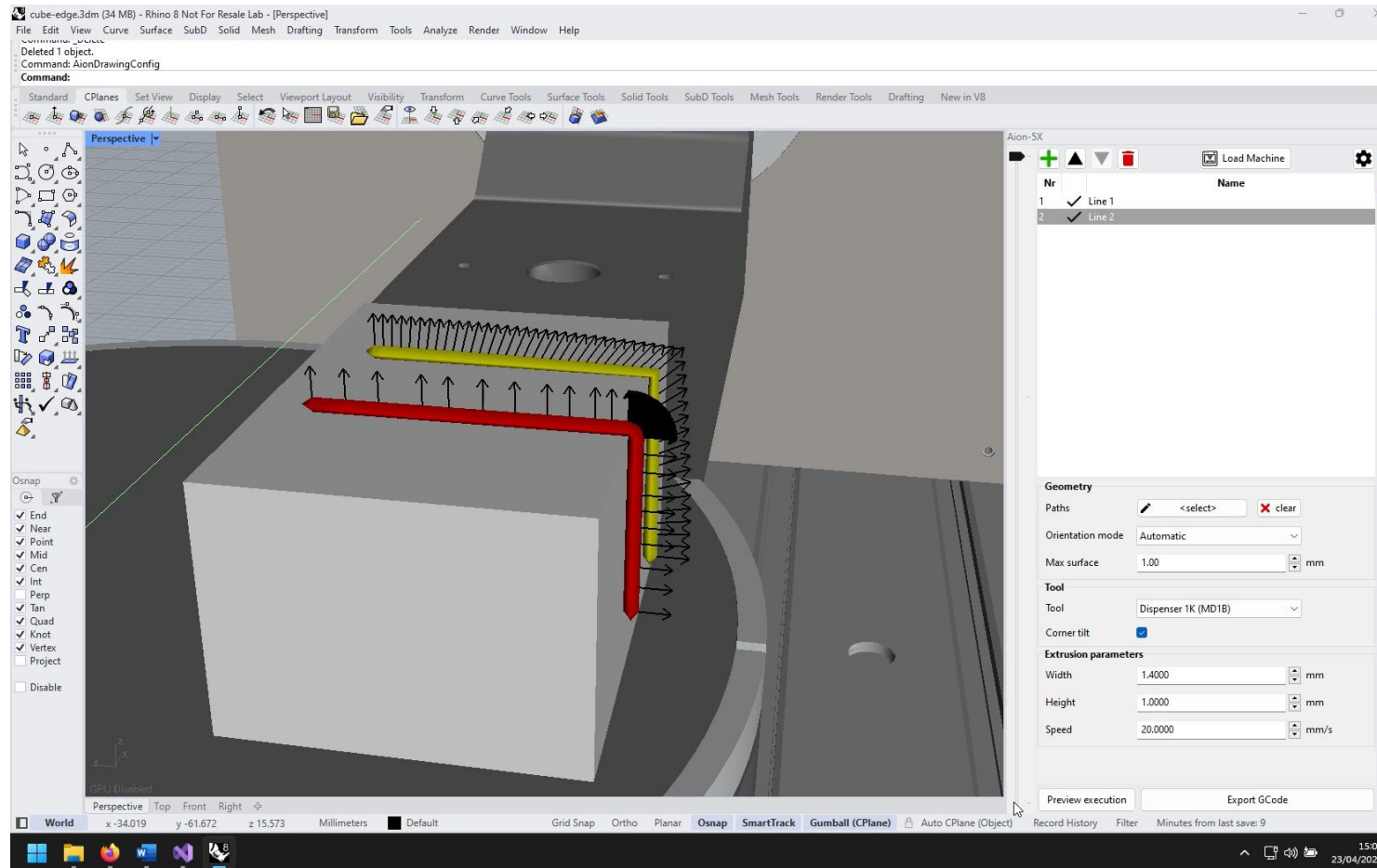
Multi-direction FFF



CAM - Manufacturing Strategies

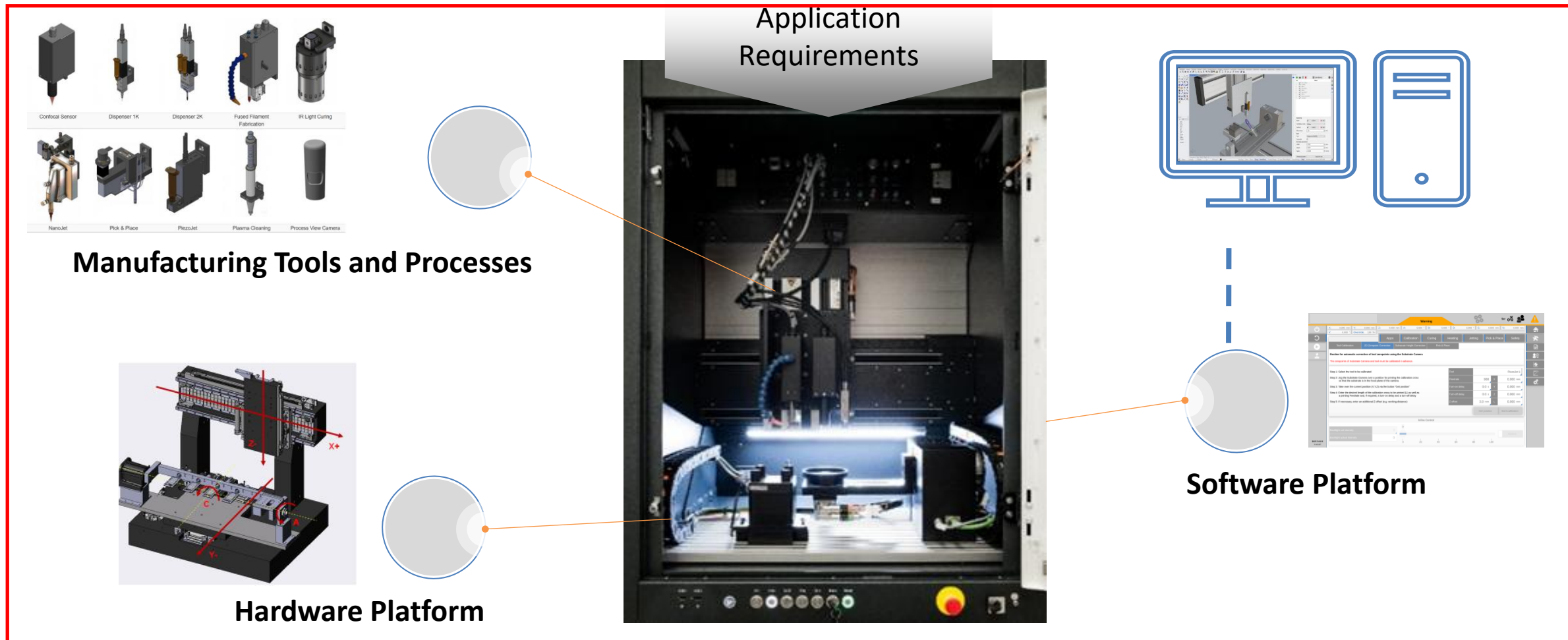


Adapted 3D Toolpath Strategies

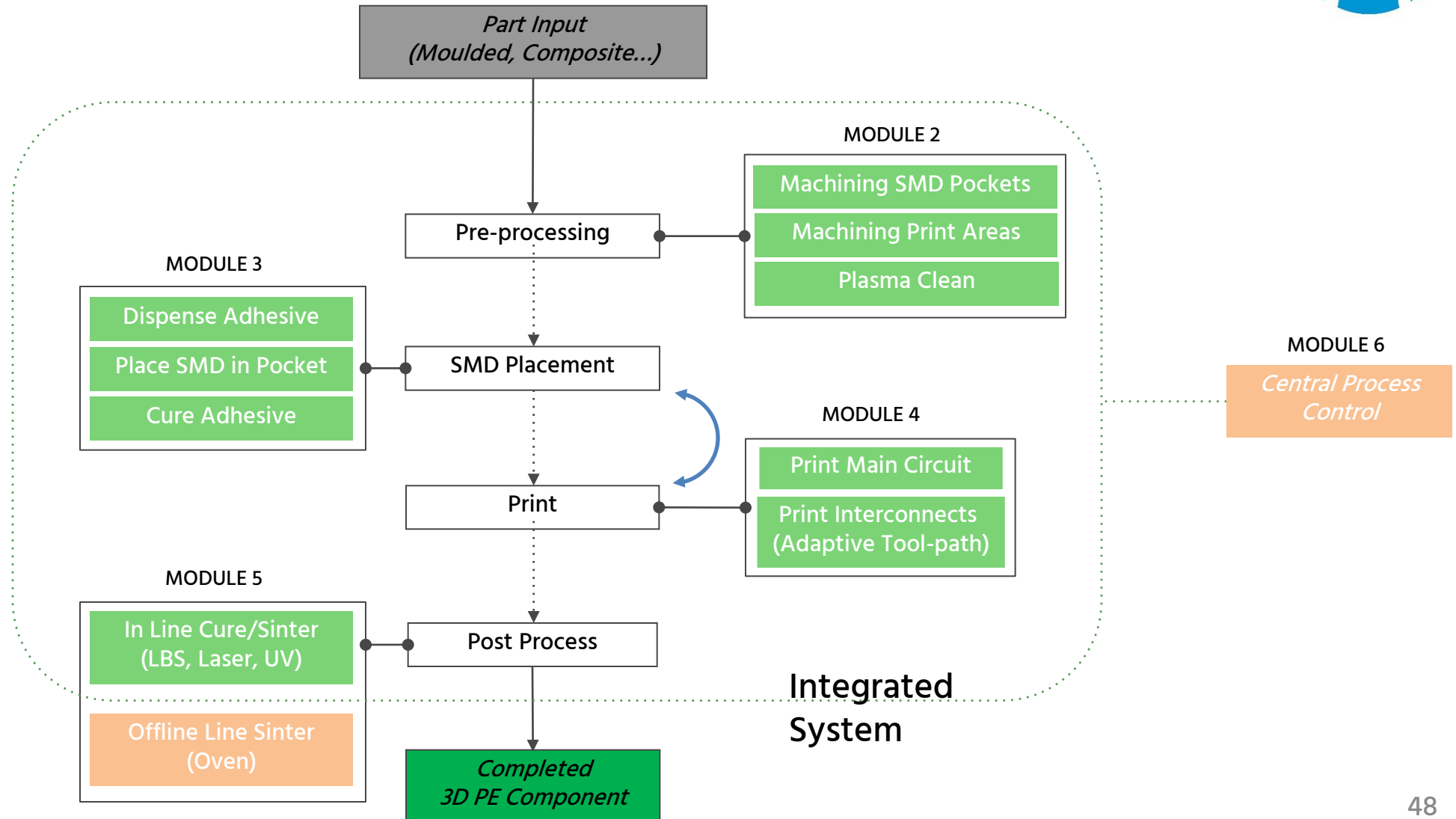


**Executable
CNC-Code
(G-Code)**

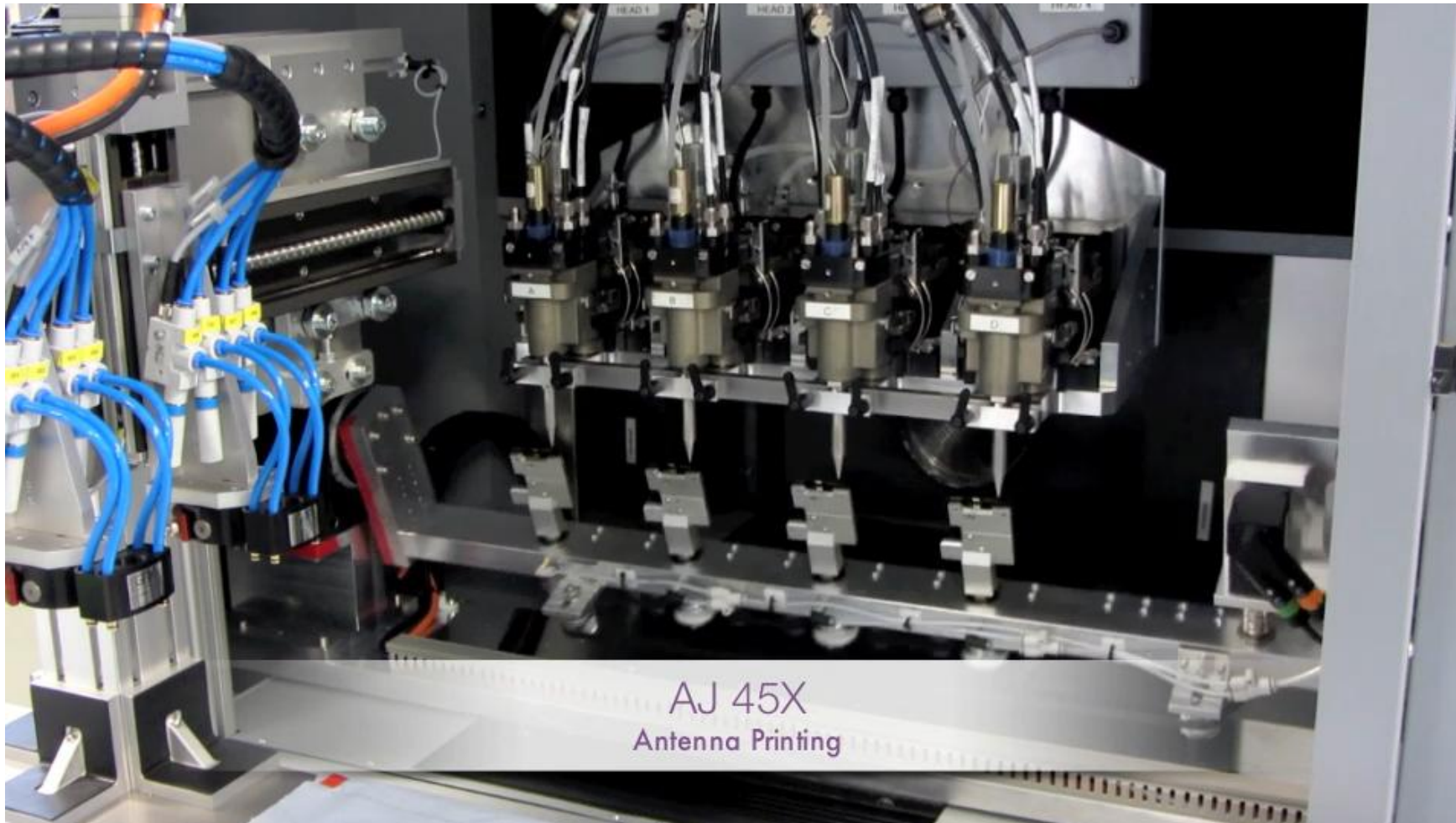
Manufacturing of 3D printed electronics using a versatile platform strategy



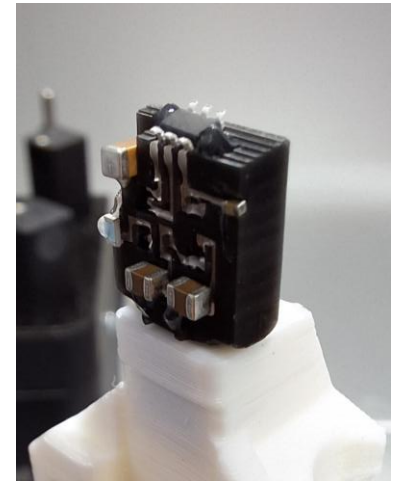
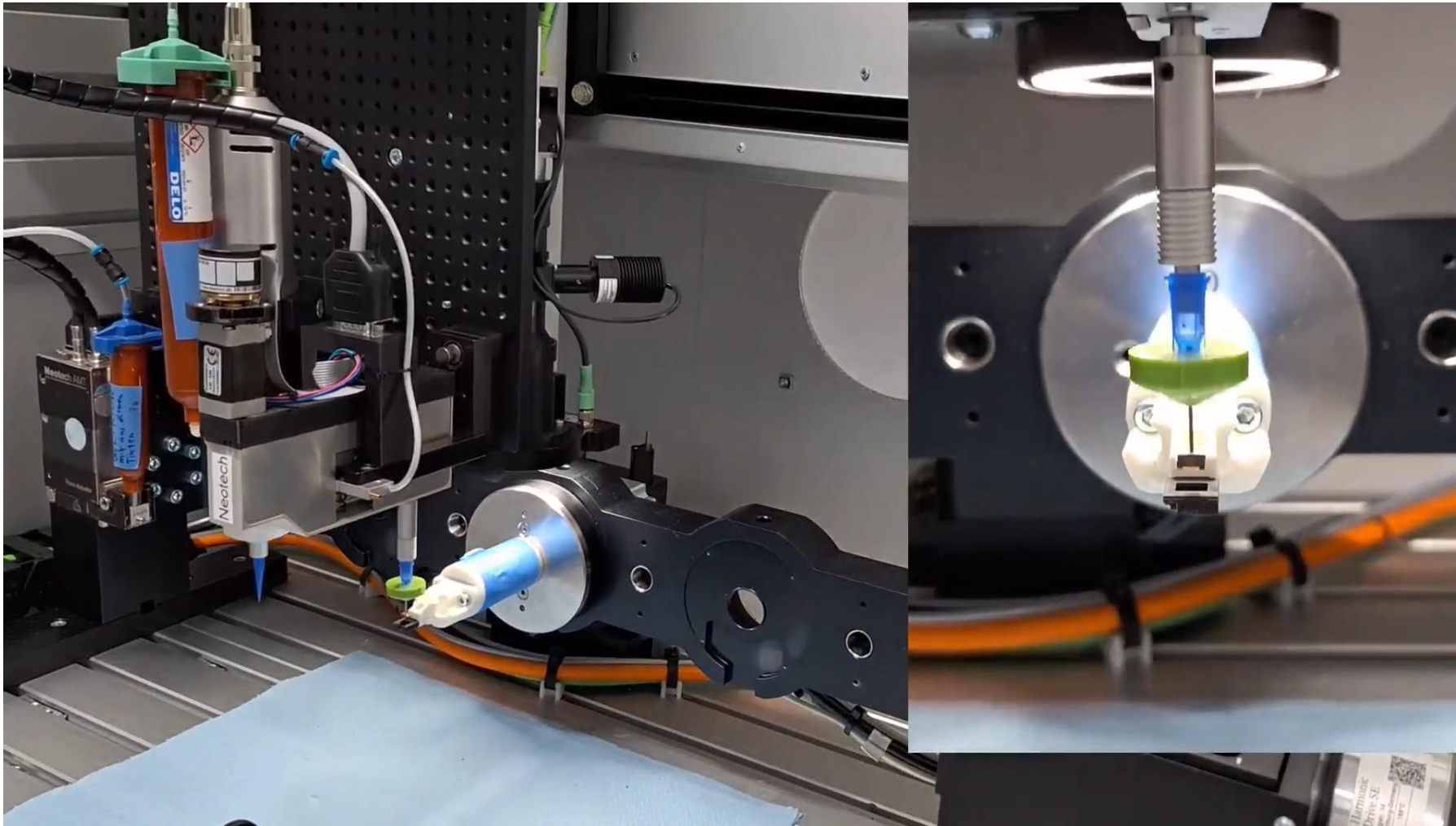
Process Route for Printing on existing 3D Substrates



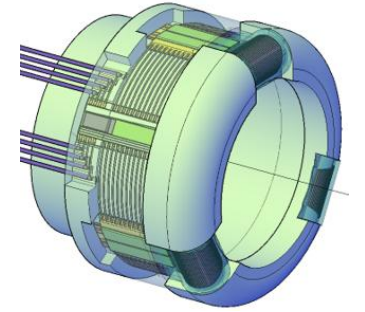
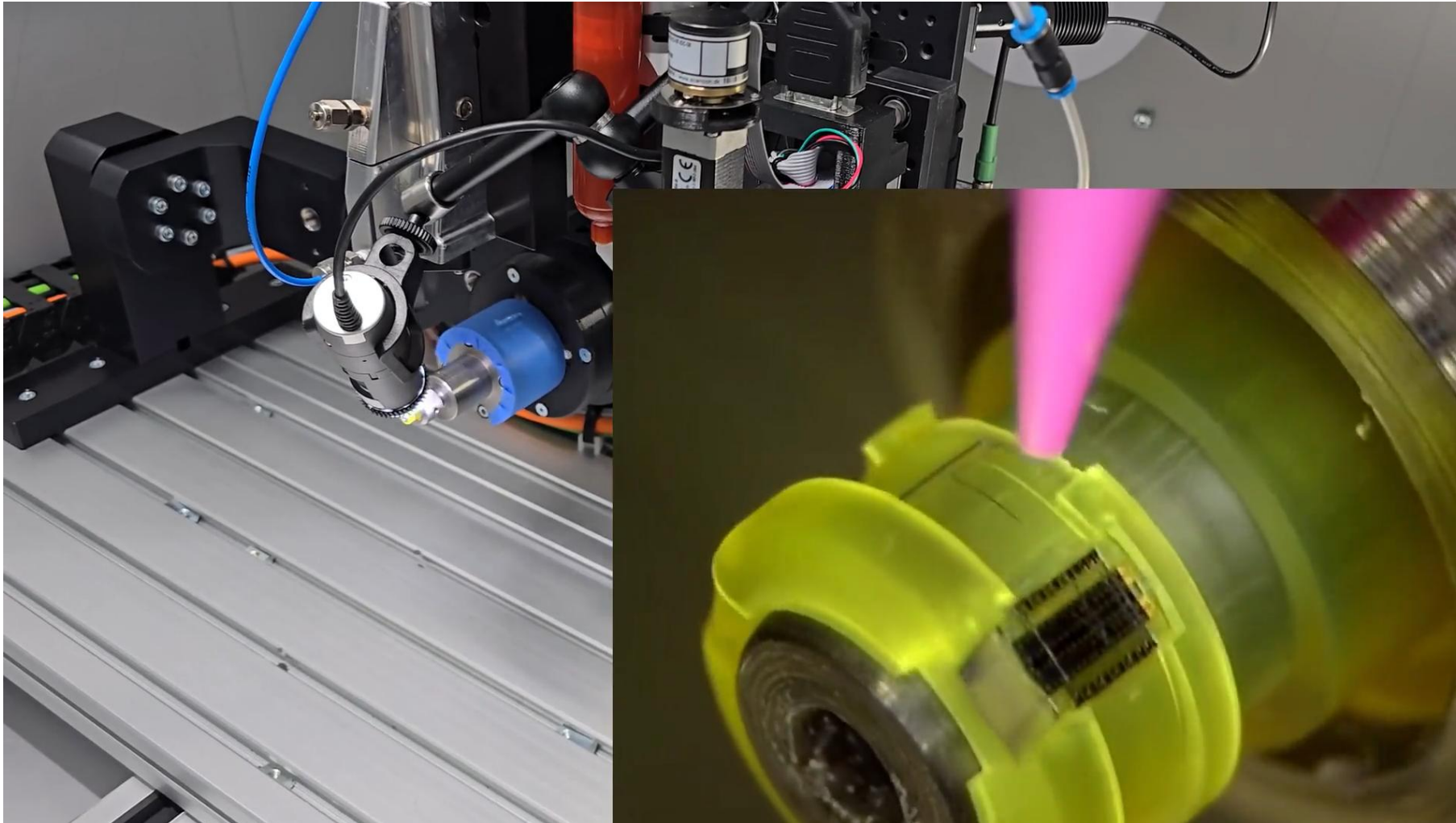
Manufacturing – Printed Antennas



Manufacturing - SPE Connector



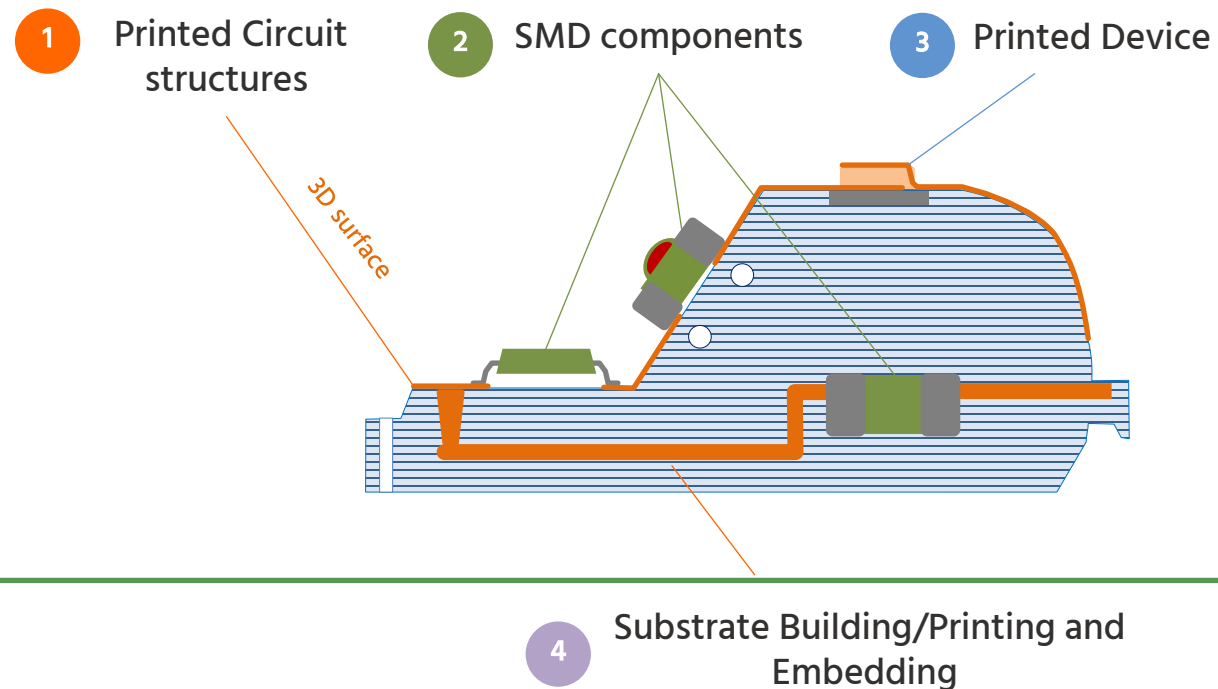
Manufacturing – ASIC Integration for Ultrasonic Camera



Fully Additive Manufacturing



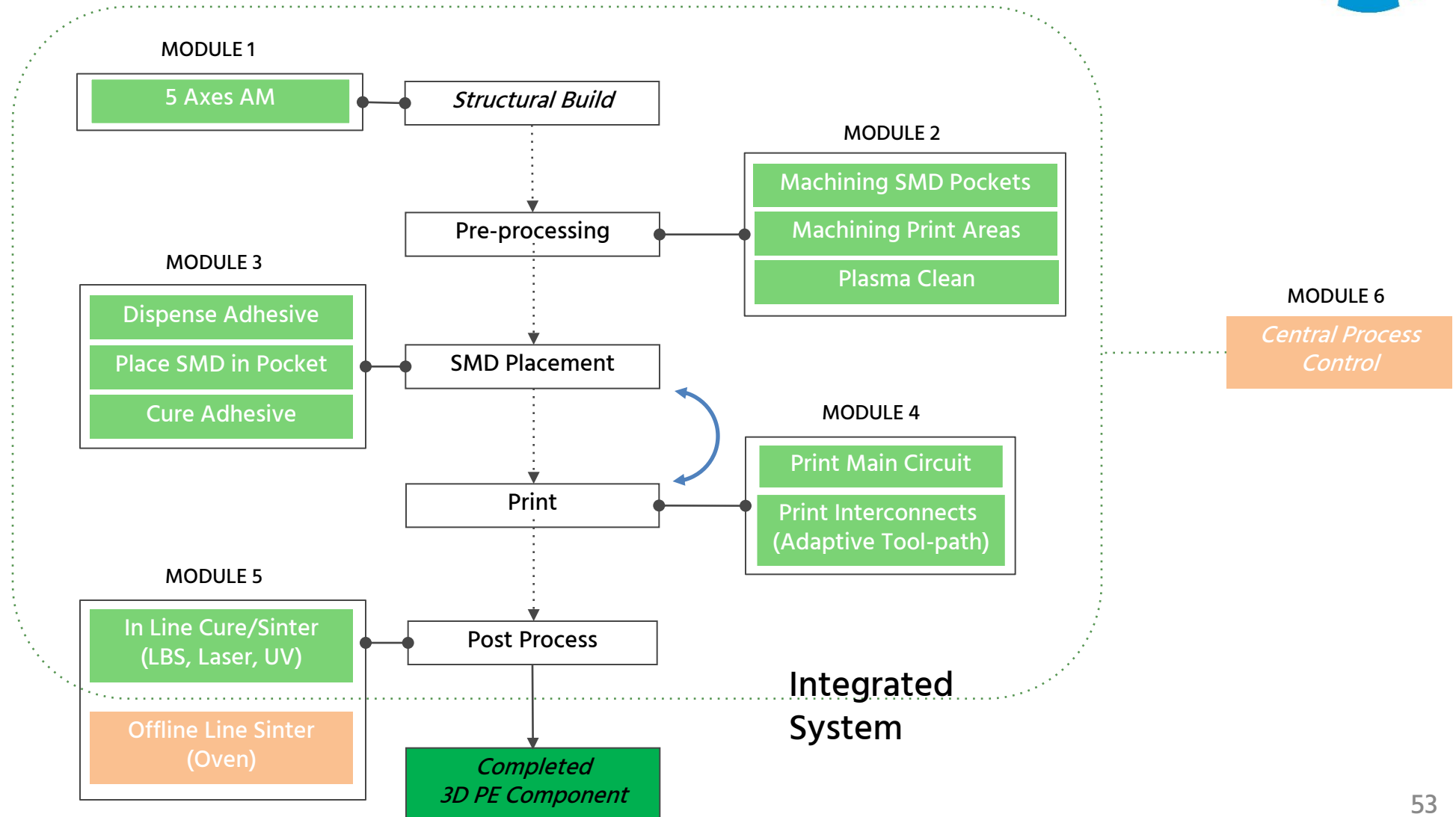
How to add electronic functionality to 3D shaped parts?



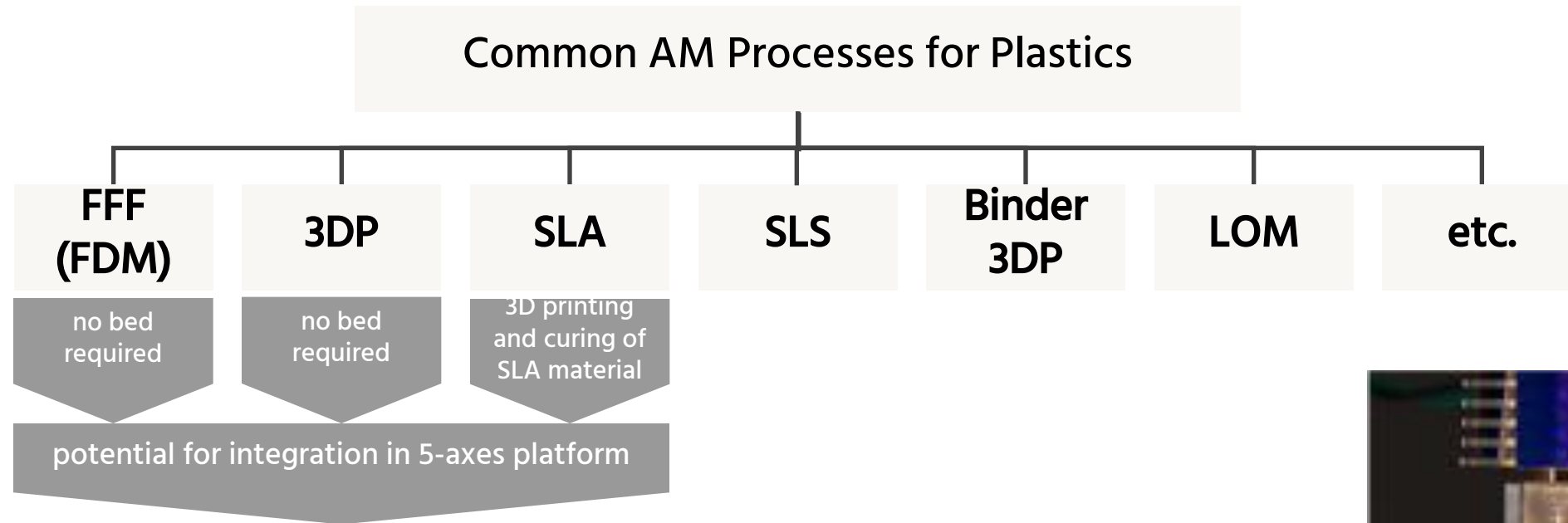
Method 1: Print on existing 3D parts (moulded, composite...)

Method 2: Fully Additive – combine 3D PE with layer-by-layer process (FDM, SLS, SLA...)

Process Route for Fully Additive Manufacturing

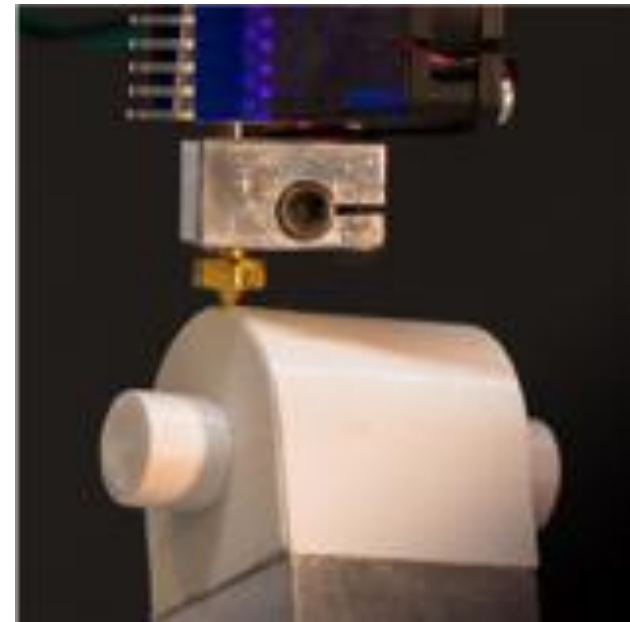


Integration of AM Processes



Benefits:

- Broad range of materials to meet product specification (in particular temperature)
- Combination of processes for increased throughput and improved surface quality
- Option to avoid support structures and anisotropic part behavior by 5-axes processing
- Integrated postprocessing of deposited materials → (Automated) Embedding of components



Application Example - 3D Printed Luminaire



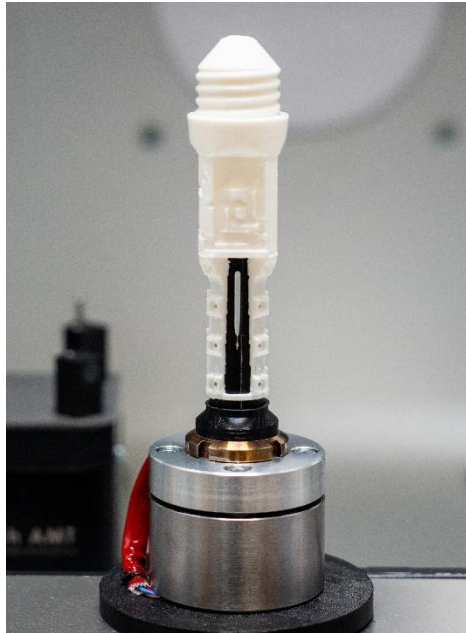
Example: Fully printed lamp with increased product sustainability:

FFF structural build



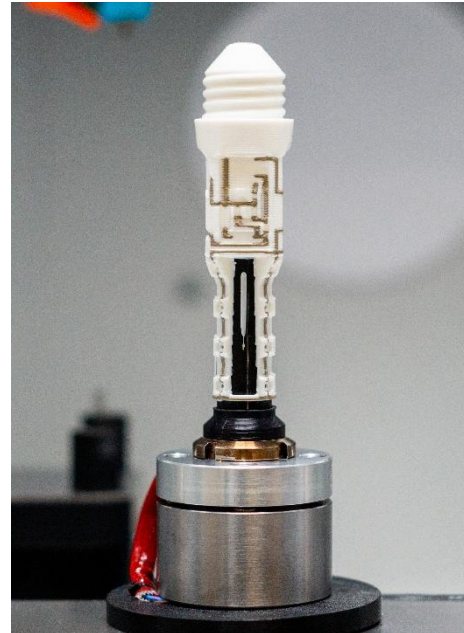
FFF on rapid
prototyping fixture

PiezoJetting of
conductive materials



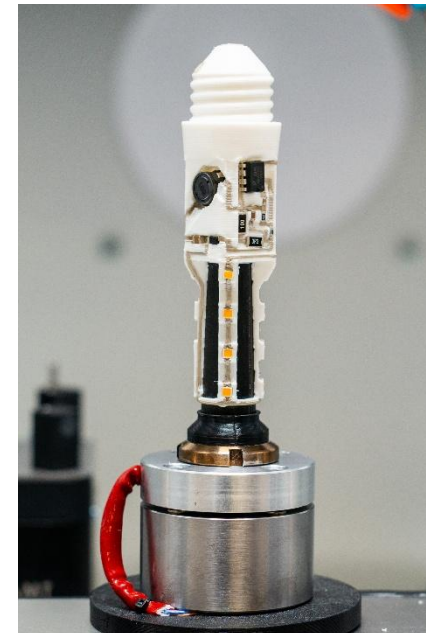
FFF printed body

Pick&Place of SMD



Ag circuit added

Pick&Place of SMD



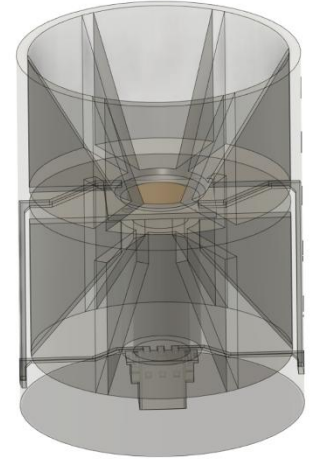
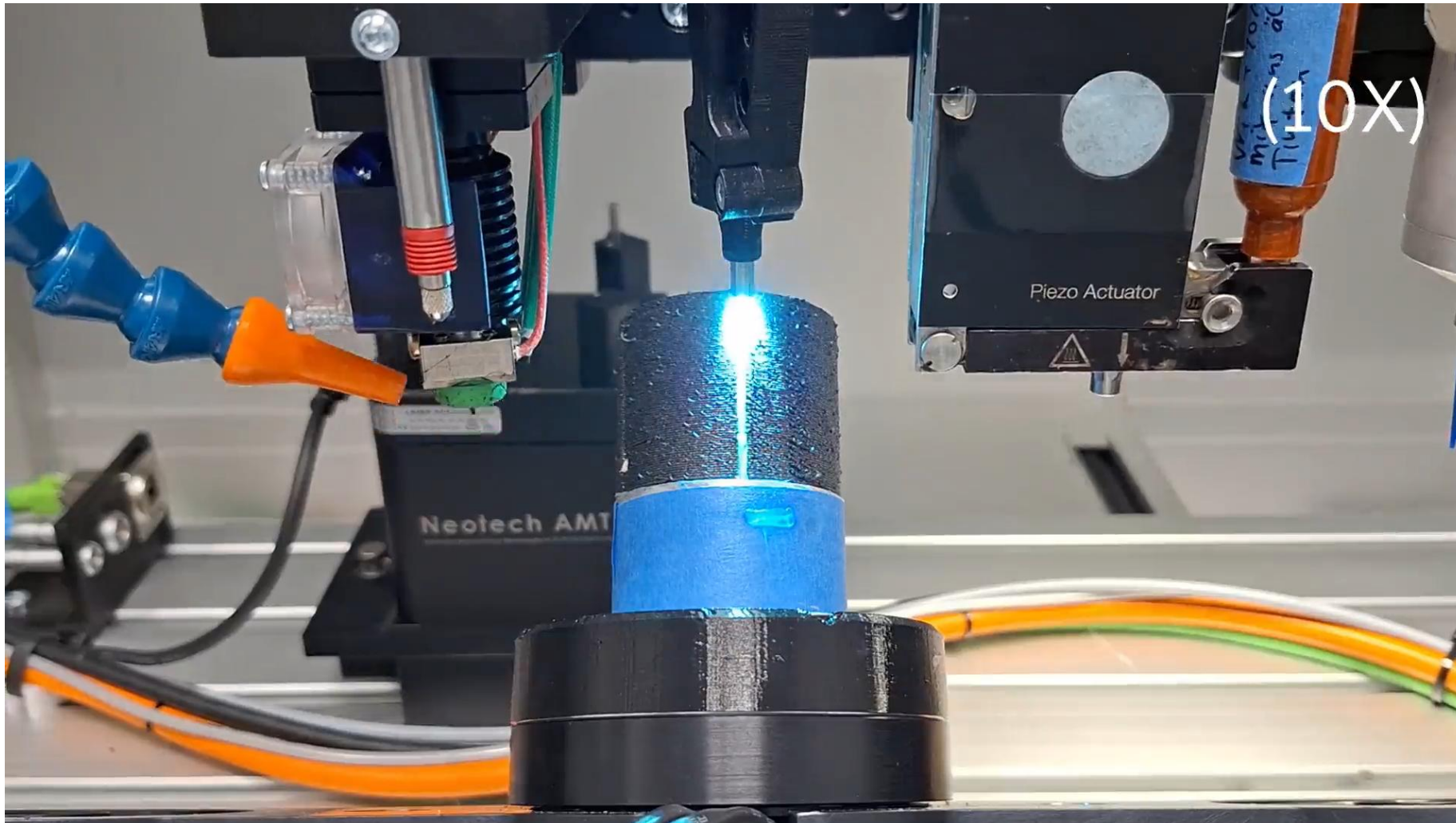
LEDs, Resistors etc.
added

FFF structural build



Printed Cover/Shade

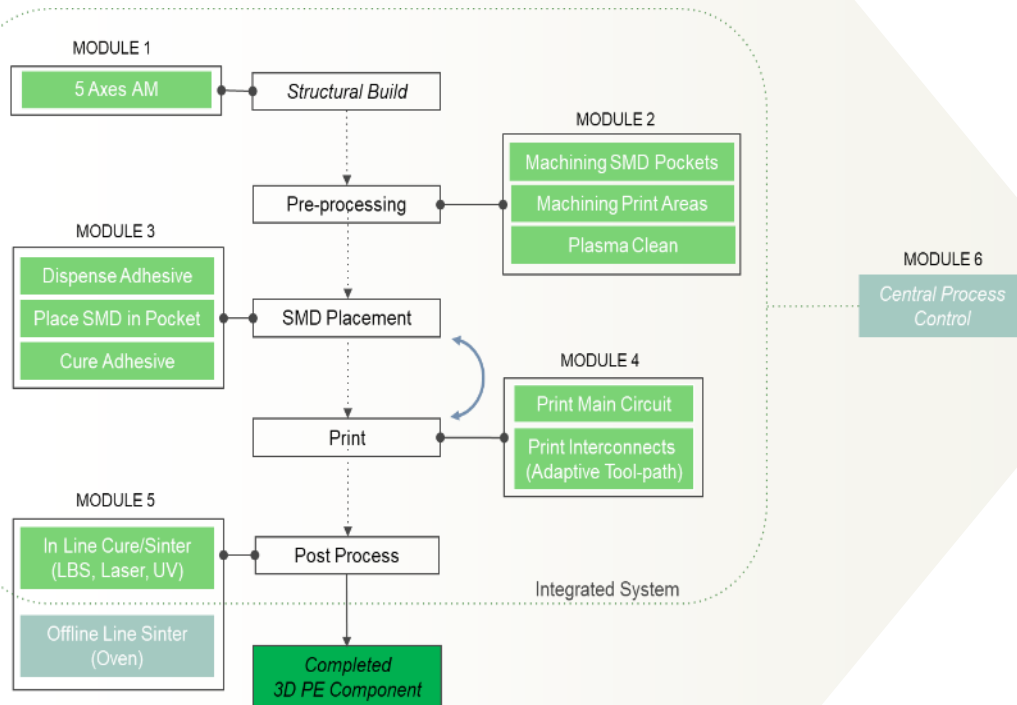
Manufacturing - Spot Light Demonstrator



Future Challenges



Process Chain:



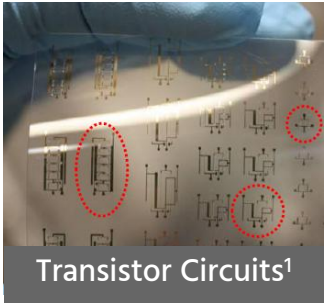
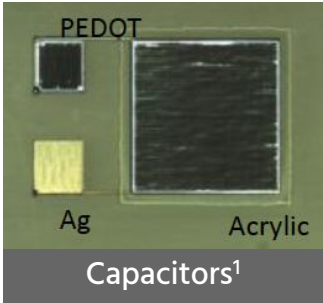
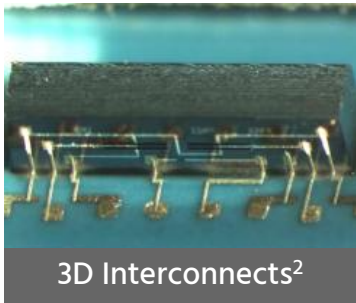
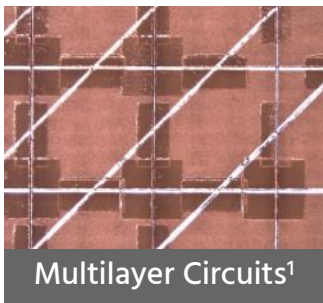
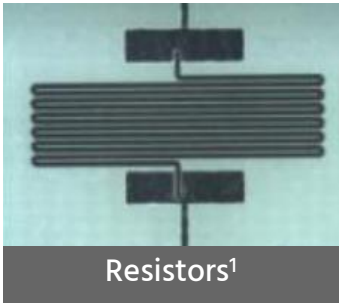
Additive Manufacturing of electronic/mechatronic systems, starting from „Zero“, faces a variety of challenges:

- **Concepts and Design** – Training, Design Guidelines and enhanced Software Tools
- **Multi-material combinations** – Development of new materials; Investigation of material processing and interactions; tests of long-term behavior according to specifications of the application
- **Sustainable Production and Products**
- Technologies and systems for **serial products** – Definition of process windows; Adapted Hardware
- **First time right production** – Process control; Integration of sensors and adapted control systems
- **Test, Maintenance and Repair** – Development of concepts and systems for test and repair; investigation of the influence on product performance
- Development of **Standards** and test criterions
- ...

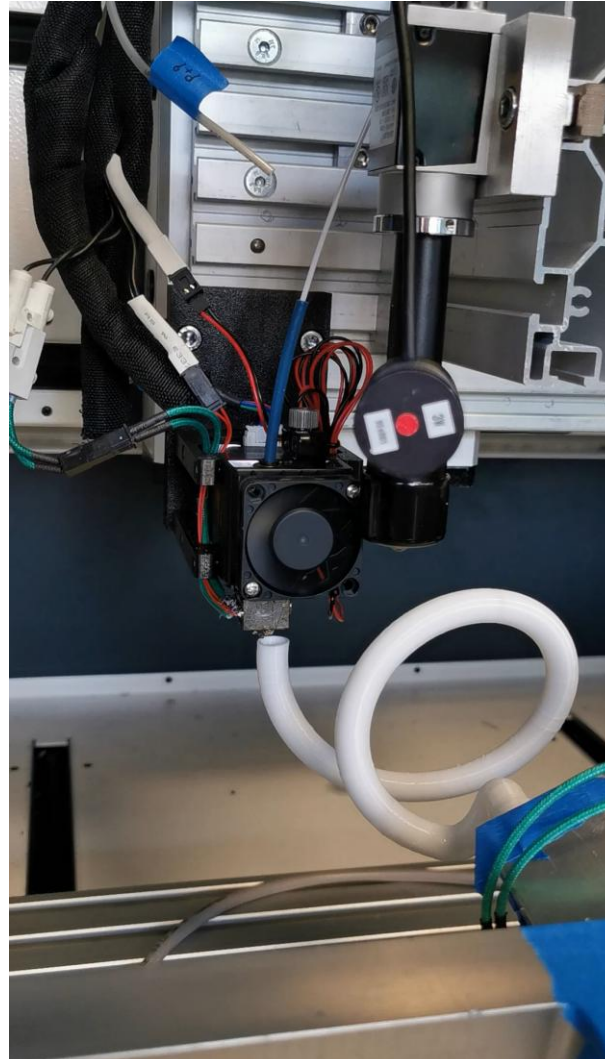
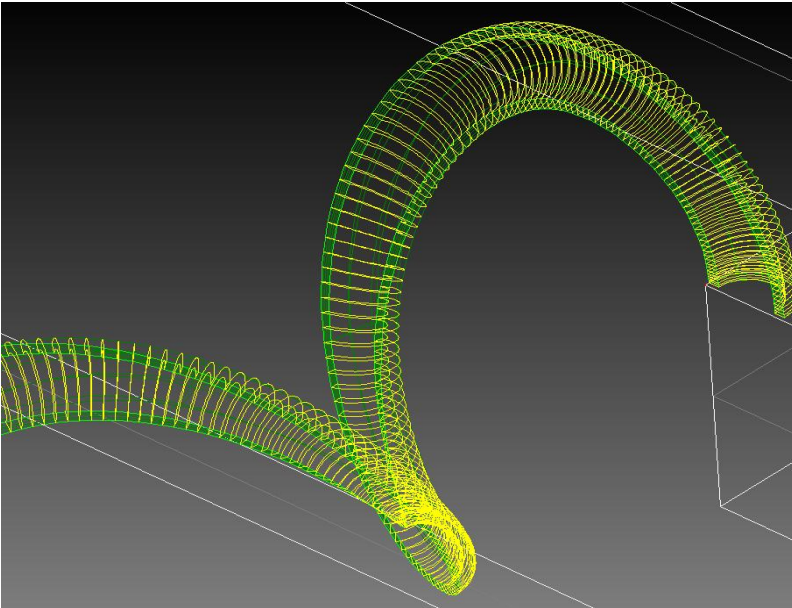
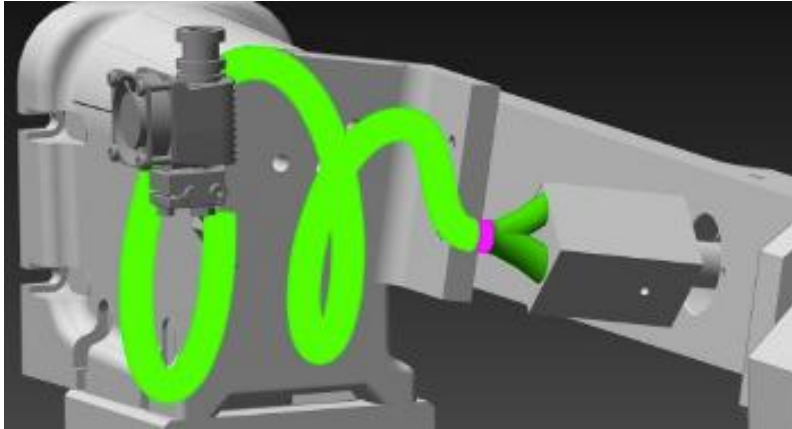
Adding Advanced Printed Structures



<i>Printed Component on 3D</i>	<i>Function</i>
Conductors	Carry Current
Antenna	Broadcast/Receive
Sensors	Input
Heater	Heat Part
Resistors	Control Current Flow
Capacitors	Filter, Charge Storage
Inductors	Filter, Transformer
Diodes	Valve
Transistors	Amplify, Switch
Memory	Information Storage
Emitters	Display Output
Power Source	Energise Circuit



Development of Multiaxis / Freeform Slicing Algorithms



Manufacturing of complex 3D shapes and e.g. tubes without supporting structures.

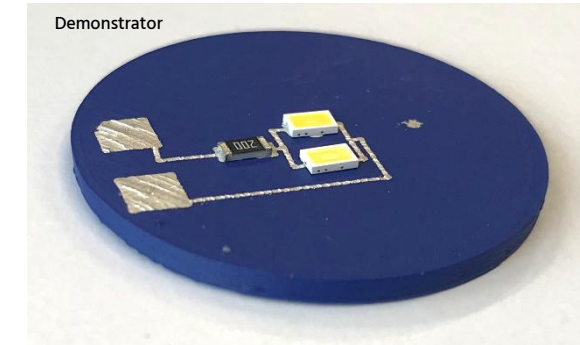
Printing of Advanced Materials (Project: PE3D)



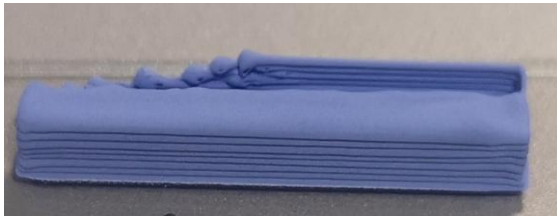
Development of Additive Manufacturing processes for LTCC ceramic substrates with integrated circuit tracks for high temperature automotive applications including HF/antenna and sensor units.

Main benefits of LTCC (Low Temperature Cofired Ceramic) substrates:

- high thermal stability and good thermal conductivity,
- low CTE → enables reliable component interconnect,
- adapted RF performance.



Dispensing of LTCC Paste



- + Low shrinkage
- + High throughput
- Pinning process using the LBS-Module needs to be combined with Dispensing
- No complex geometries
- Tailormade pastes

Extrusion of LTCC Filaments



- + Near Net Shape
- + 5-axis manufacturing
- High shrinkage rate during firing
- Tailormade filaments required



Post-Processing: CNC Milling

CNC Milling of green body to improve geometry and surface quality:



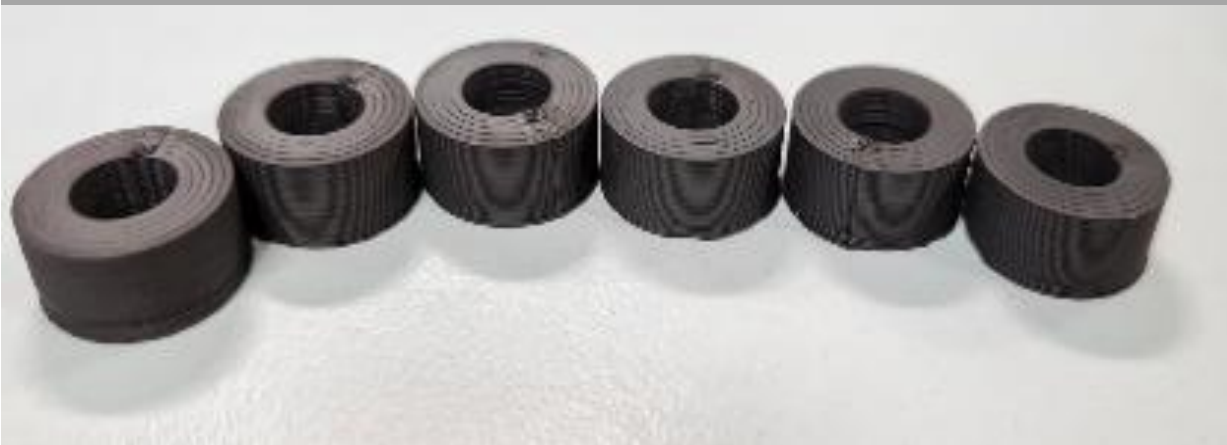
Printing of Advanced Materials (Project: MEPOFERRI)



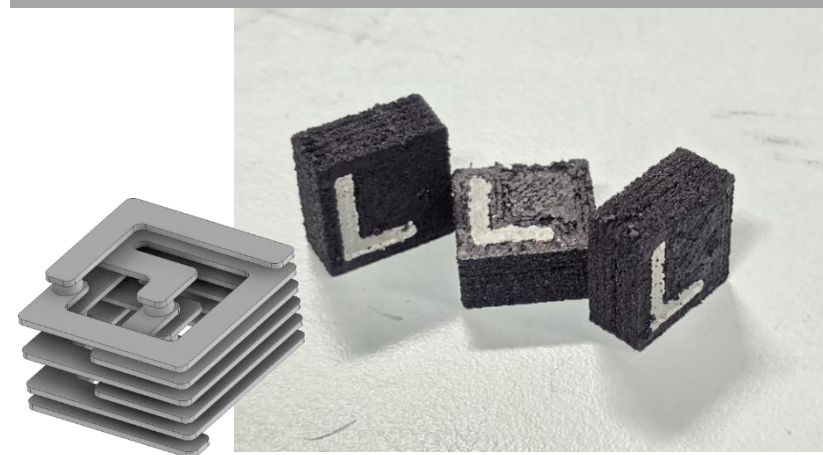
Aim: Economical manufacturing of customised ferrite parts at a high Freedom of Design.

- Development of new material combinations
- Additive Manufacturing techniques for processing highly filled metal/ferrite thermoplastics
 - Fused Filament Fabrication
 - Fused Granulate Fabrication
 - Dispensing
- Process routes in combination with pre-/postsintering

Additive Manufactured Ferrite Cores



Embedded Printed Coils



ADDIRA – AM for Automotive Radar Sensors

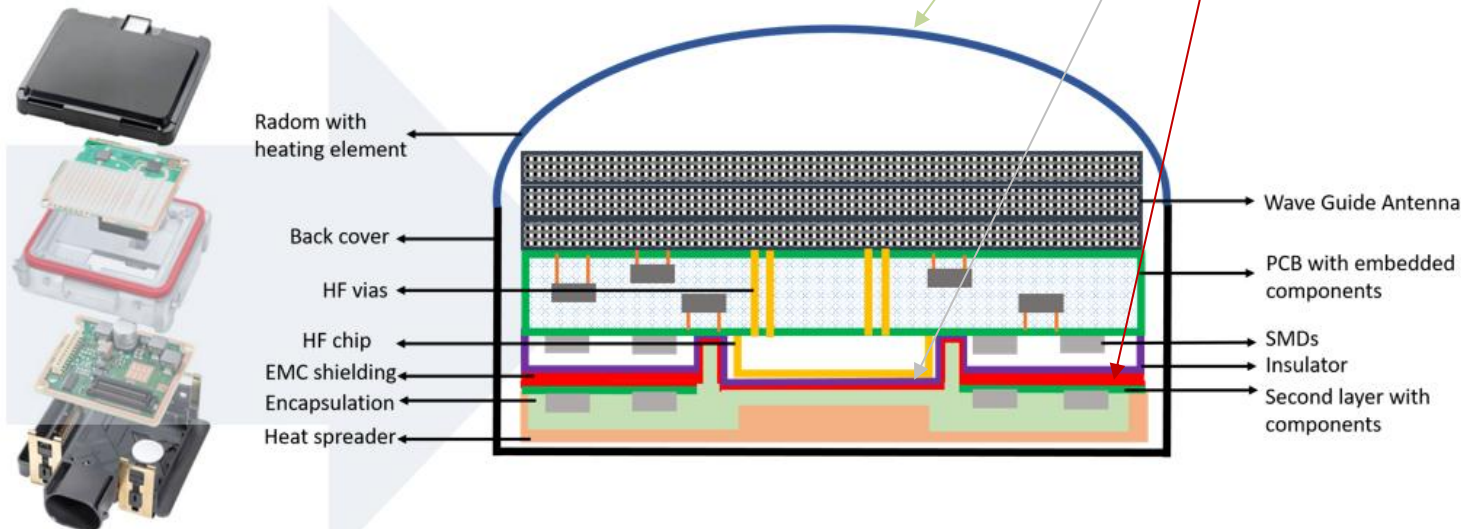


Aim:

Increase passenger safety by improving the performance of Automotive Radar Sensors.

Additive Manufactured structures are developed to:

- Improve heating performance for reducing the de-icing time of the sensor
- To substitute metal EMC shielding by printed conductive layers



Process Development

Liquid Metal Jetting

- Using Liquid Metal Jetting and FFF to manufacture a Radome with embedded heater

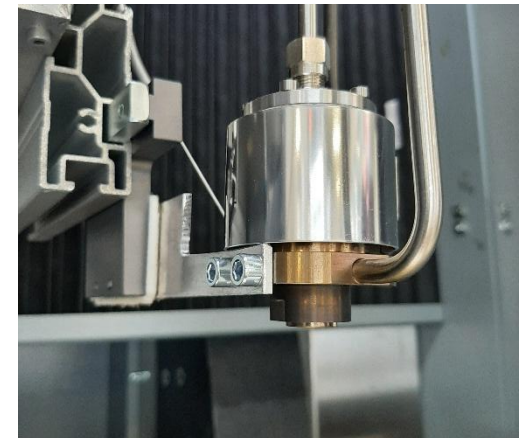
Piezo Jet

- Printing of copper-based sinter materials for Die Attach.

Spray Module

- Spray Coating of conductive materials on the sensor housing for EMS

Printhead (prototype):

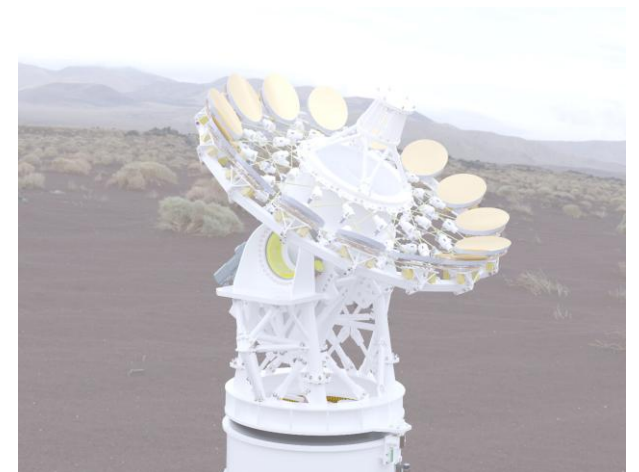


EIC research project „Live Mirror“ - Overview



A long-term scientific valorization towards a disruptive concept of telescopes, making use of **Live-Mirror developments** dedicated to the detection and imaging of exoplanets.

Short-term application:
e.g. wireless optical communications



EIC research project „Live Mirror“ – Terpo4 Printing



Active/Living Mirror
Optical Surface

AM actuators between optical and
reaction surface

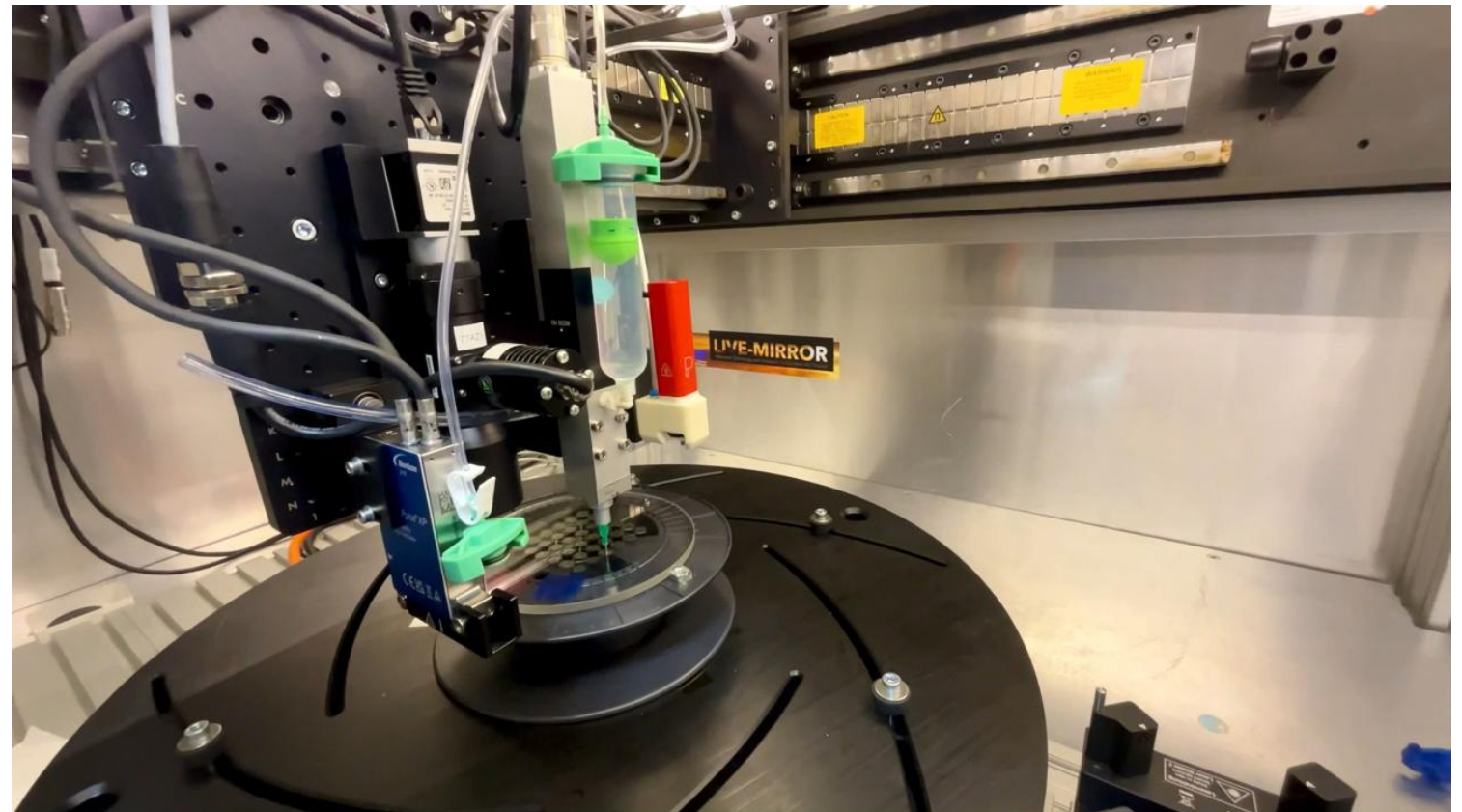
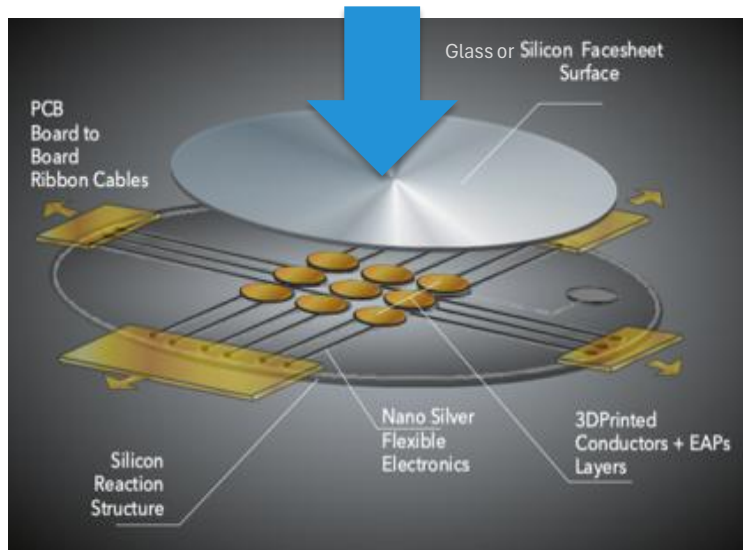
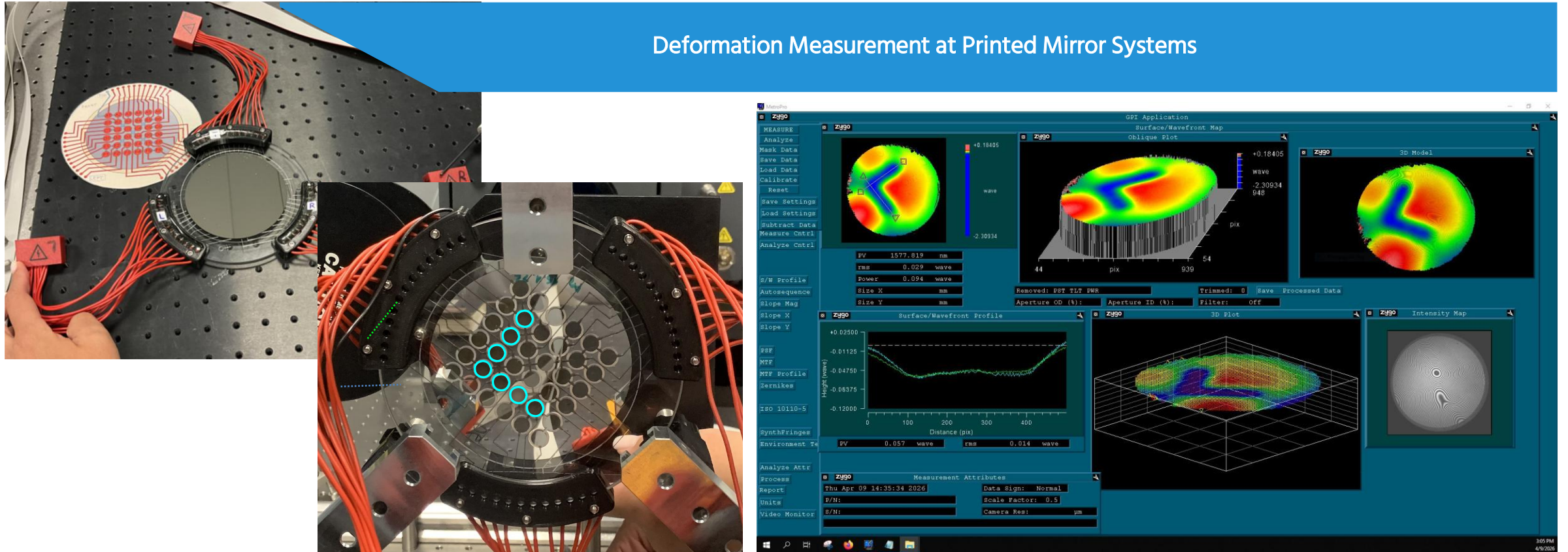


Image credits: Live Mirror consortium

Multilayer Multimaterial Printing to manufacture interconnected actuators directly on curved glass elements

EIC research project „Live Mirror“ – Characterization



Multilayer Multimaterial Printing to manufacture interconnected actuators directly on curved glass elements

Print Project: Medical Application



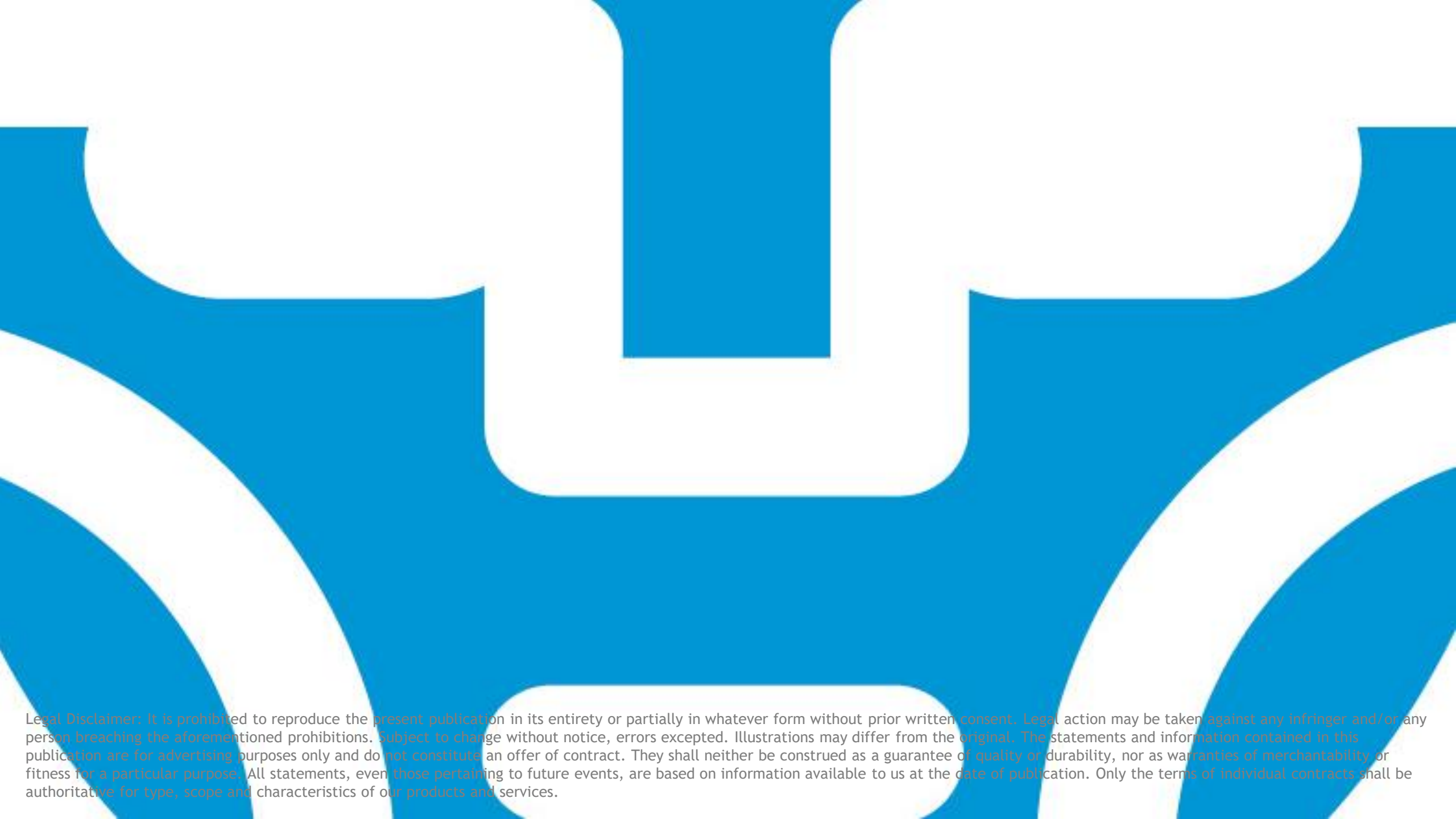
15.06.2026: Print Trials ongoing during Lab Tour

Summary



The intention of the presentation is to provide an overview on following topics:

- 5 Axes motion systems are combined with contactless printing technology to apply functional materials onto 3D substrates to create 3D Mechatronic Devices (3D-MID).
- An adapted software tool is available for designing 3D print toolpaths for the various print technologies.
- Currently the focus of the applications is on “simple” printed electronic functions applied on substrates ranging in size from a few Millimetres up to approx. 2m x 1m.
- For the manufacturing of hybrid electronics, a Pick&Place module and a camera based system control for adaptive toolpaths can be integrated.
- Technologies adapted from the Additive Manufacturing will enable a fully additive manufacturing 3D electronic devices “starting from zero”, even in lot size 1. Process routes are currently under development.
- Besides the processes and process routes the development of fully additive manufactured mechatronic devices faces additional challenges along the entire life cycle, starting from the design to the End-of-Service and Recycling. Massive progress has been and is being made, not least through publicly funded research projects.



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